Chapter 2

PLC Hardware Components

2.1

The I/O Section

The *input/output (I/O)* section of a PLC is the section to which all field devices are connected and provides the interface between them and the CPU.

Inputs

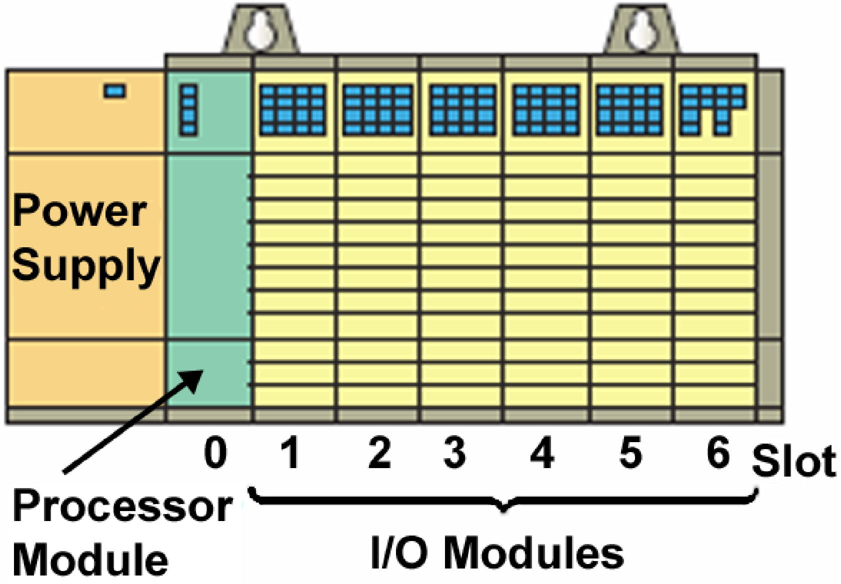
Input/output arrangements are built into a fixed PLC.

Outputs

I/O Modules

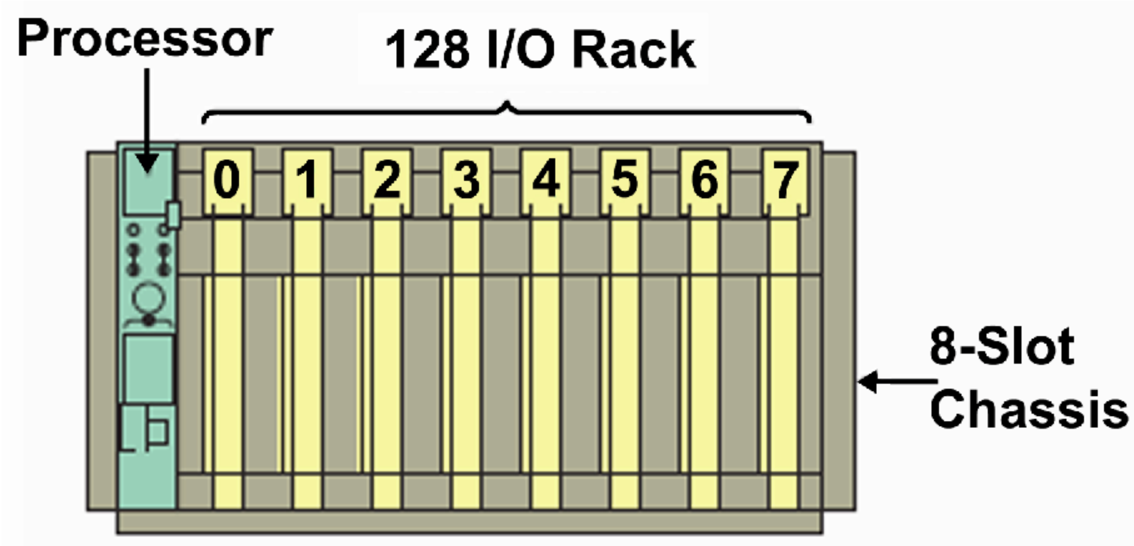
Modular types use external I/O modules that plug into the PLC.

Rack-based I/O section made up of individual I/O modules



Input interface modules ***accept signals*** **from the machine or process devices and convert them into signals that can be used by the controller.**

Output interface modules ***convert controller signals*** **into external signals used to control the machine or process.**



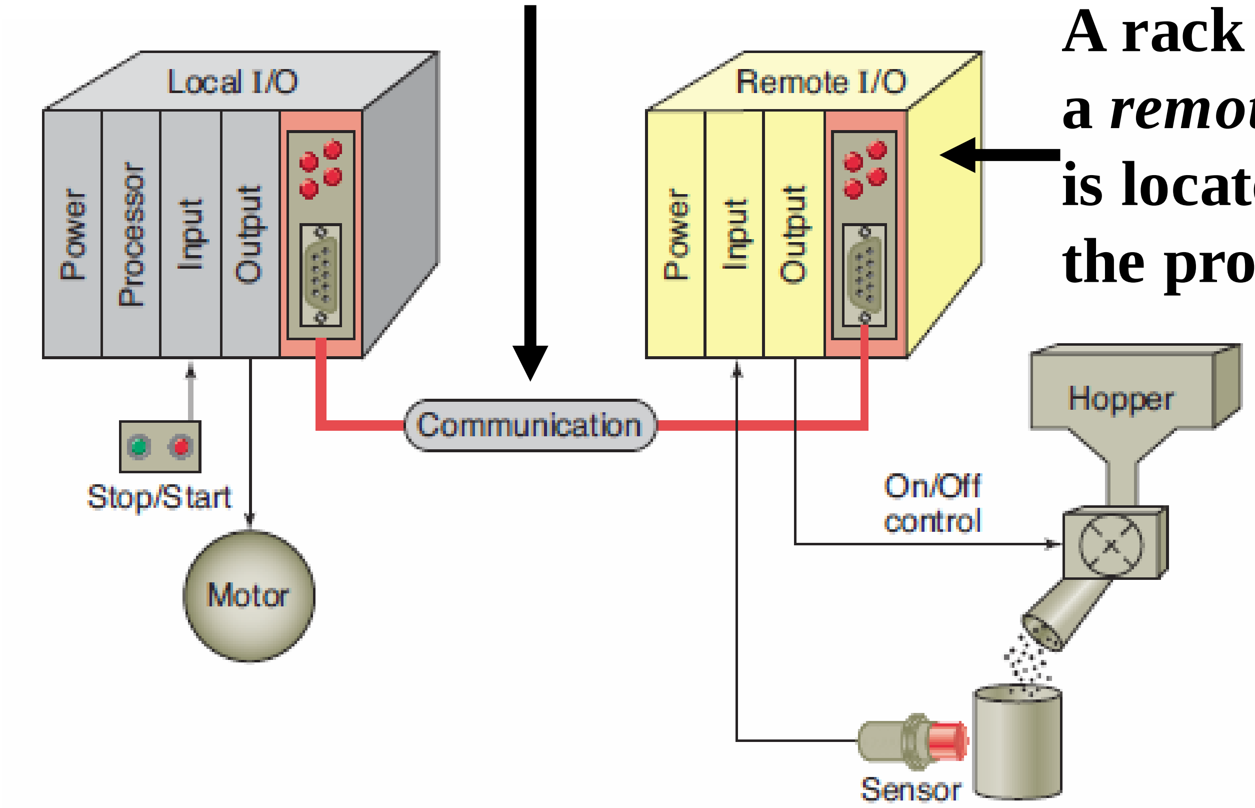
The hardware assembly that houses I/O modules, processor modules, and power supplies is referred to as the *chassis*.

A *logical rack* is an addressable unit consisting of 128 input points and 128 output points.

The ability to locate the I/O modules near the field devices minimizes the amount of wiring required.

The remote rack is linked to the local rack through a pair of *communications modules*.

A rack is referred to as a *remote* rack when it is located away from the processor module.



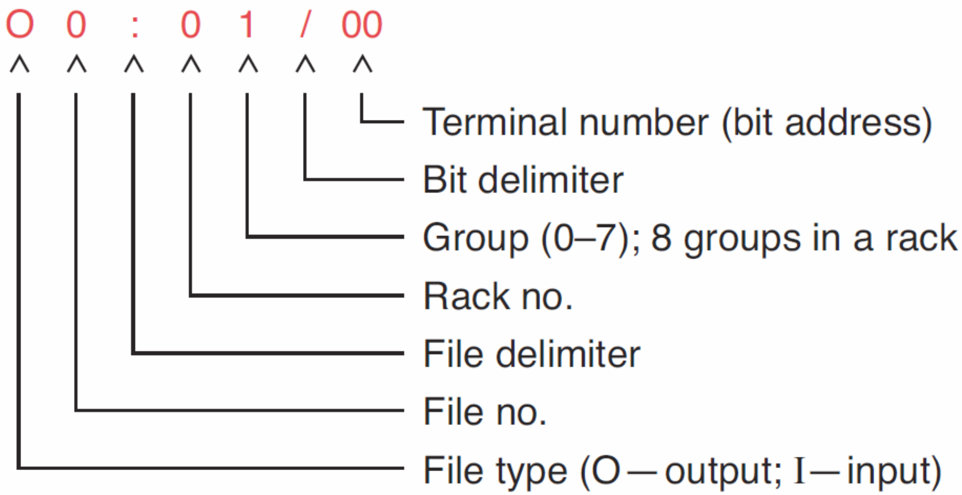
The PLC's memory system stores information about the status of all the inputs and outputs. To keep track of all this information, it uses a system called *addressing*.

- **An *address* is a label or number that indicates where a certain piece of information is located in a PLC's memory.**

- ***Rack/slot-based* addressing schemes are used with Allen-Bradley PLC-5 and SLC 500 controllers,**

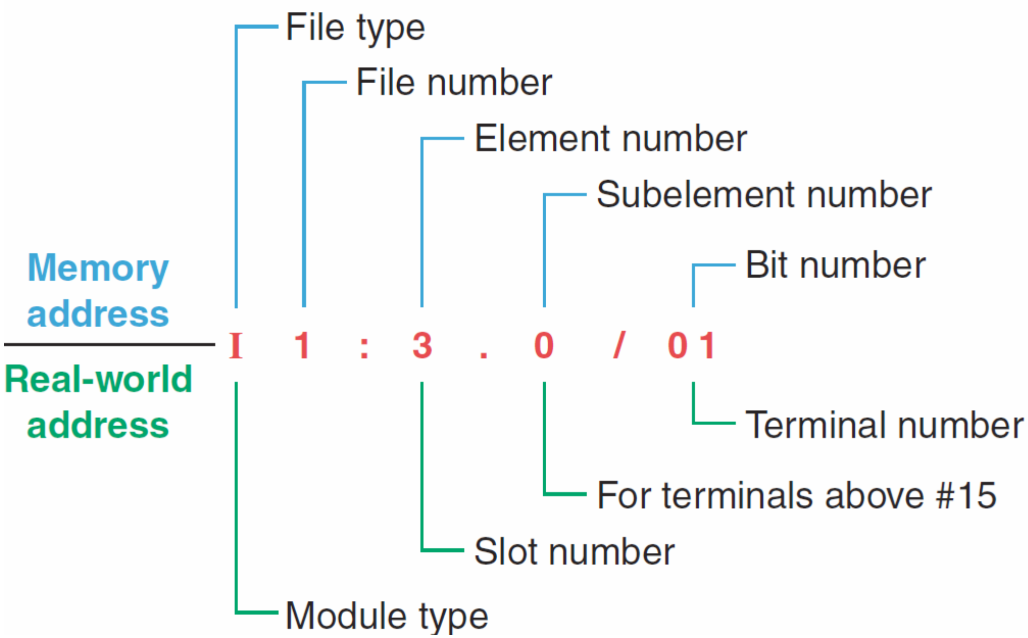
- ***Tag-based* addressing is used with Allen-Bradley ControlLogix controllers.**

PLC-5 *rack/slot-based* addressing format.



I1:27/17 - Input, file 1, rack 2, group 7, bit 17
O0:34/07 - Output, file 0, rack 3, group 4, bit 7
I1:0/0 - Input, file 1, rack 0, group 0, bit 0
O0:1/1- Output, file 0, rack 0, group 1, bit 1

SLC 500 *rack/slot-based* addressing format.



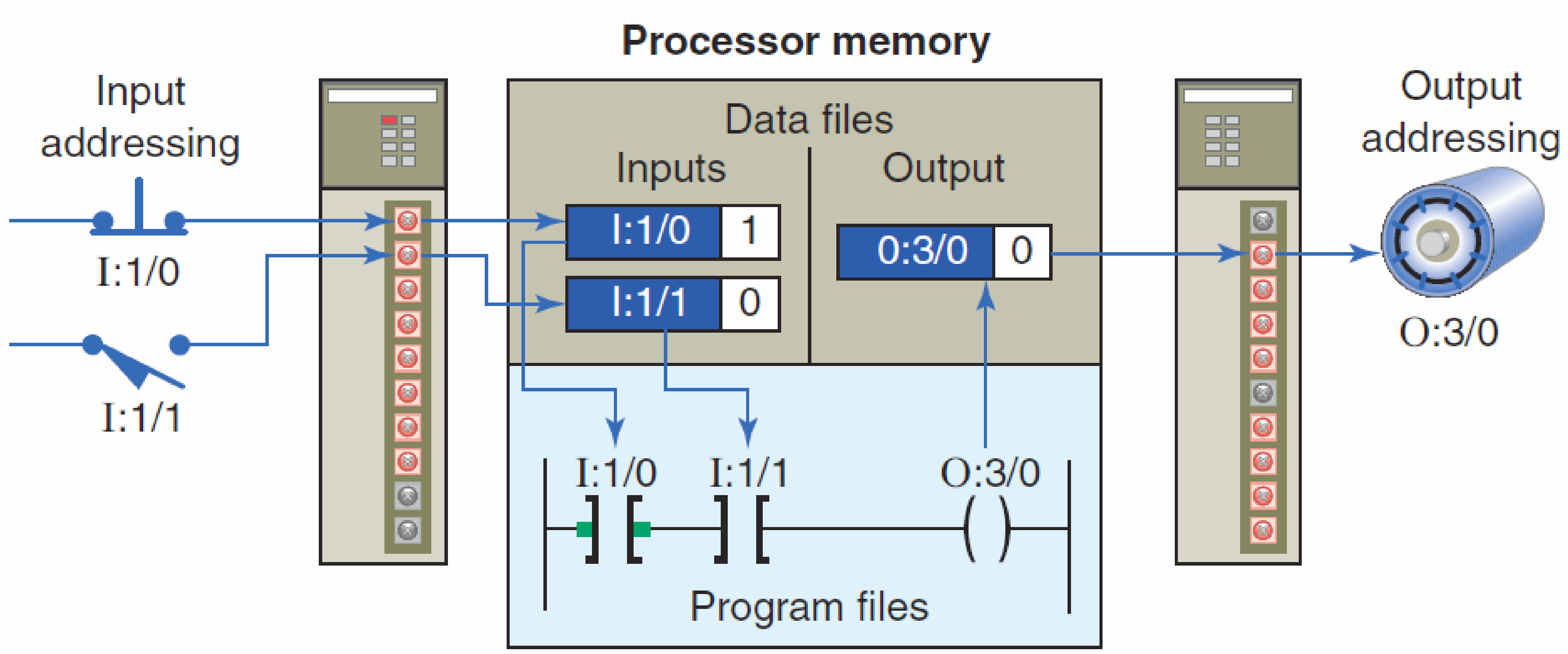
O:4/15 - Output module in slot 4, terminal 15

I:3/8 - Input module in slot 3, terminal 8

O:6.0 - Output module, slot 6

I:5.0 - Input module, slot 5

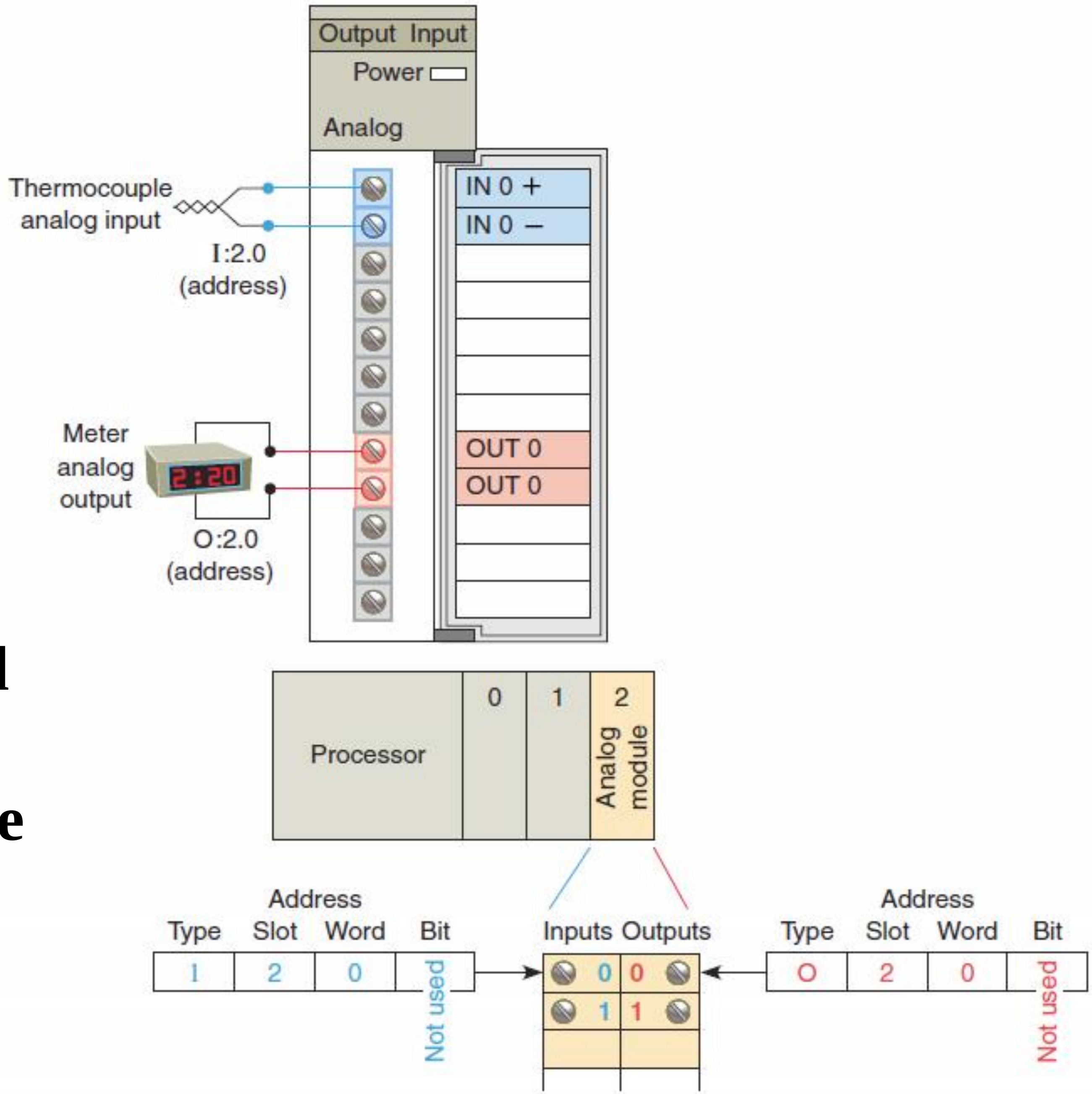
Every input and output device connected to a *discrete I/O module* is addressed to a specific *bit* in the PLC's memory.



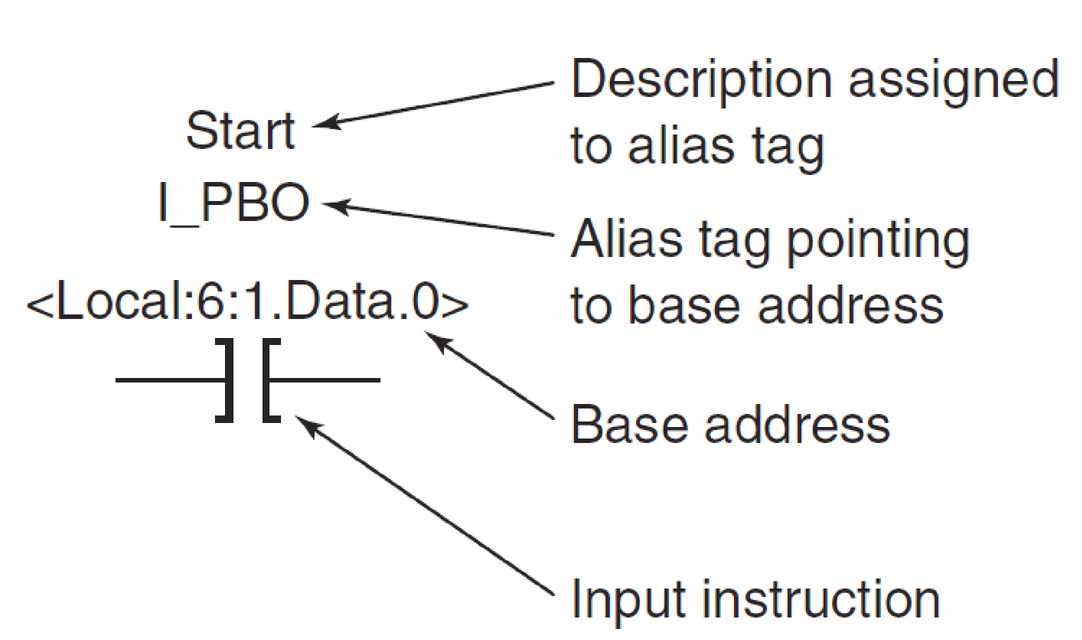
A bit is a binary digit that can be either 1 or 0.

Analog I/O modules use a *word* addressing format.

The bit part of the address is usually not used but can be addressed by the programmer if necessary.

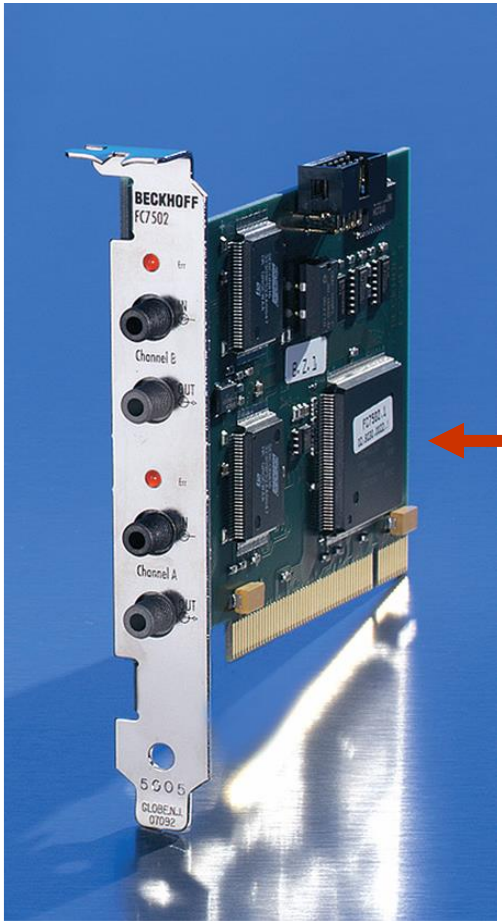


Allen-Bradley ControlLogix controllers use a *tag-based* addressing format.



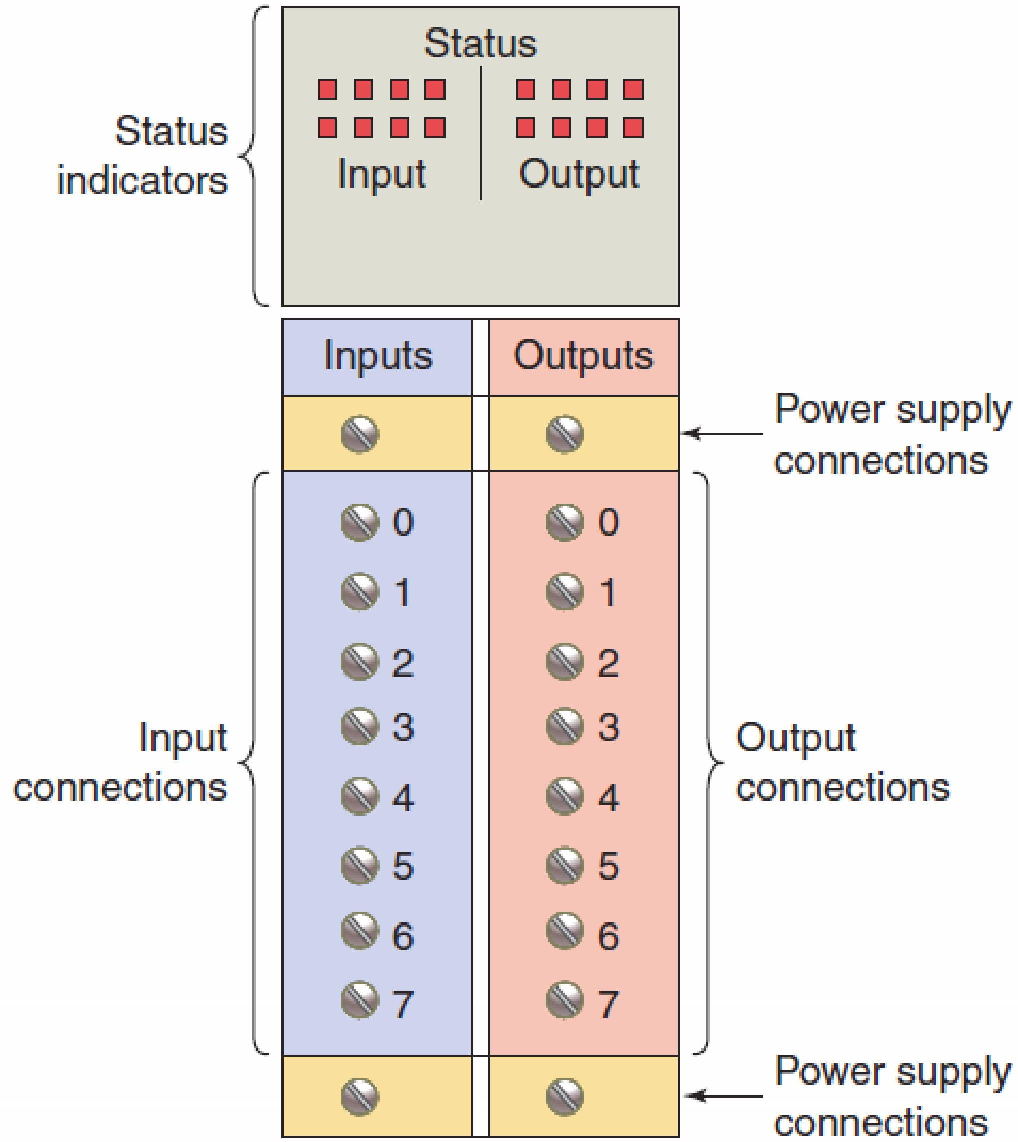
- ➢**Instead of a fixed numeric format, a *tag* (alphanumeric name) is used to address data.**
- ➢**The field devices are assigned tag names when the PLC ladder logic program is developed.**

PC-based **control runs on personal or industrial hardened computers. Also known as soft PLCs, they simulate the functions of a PLC on a PC,**

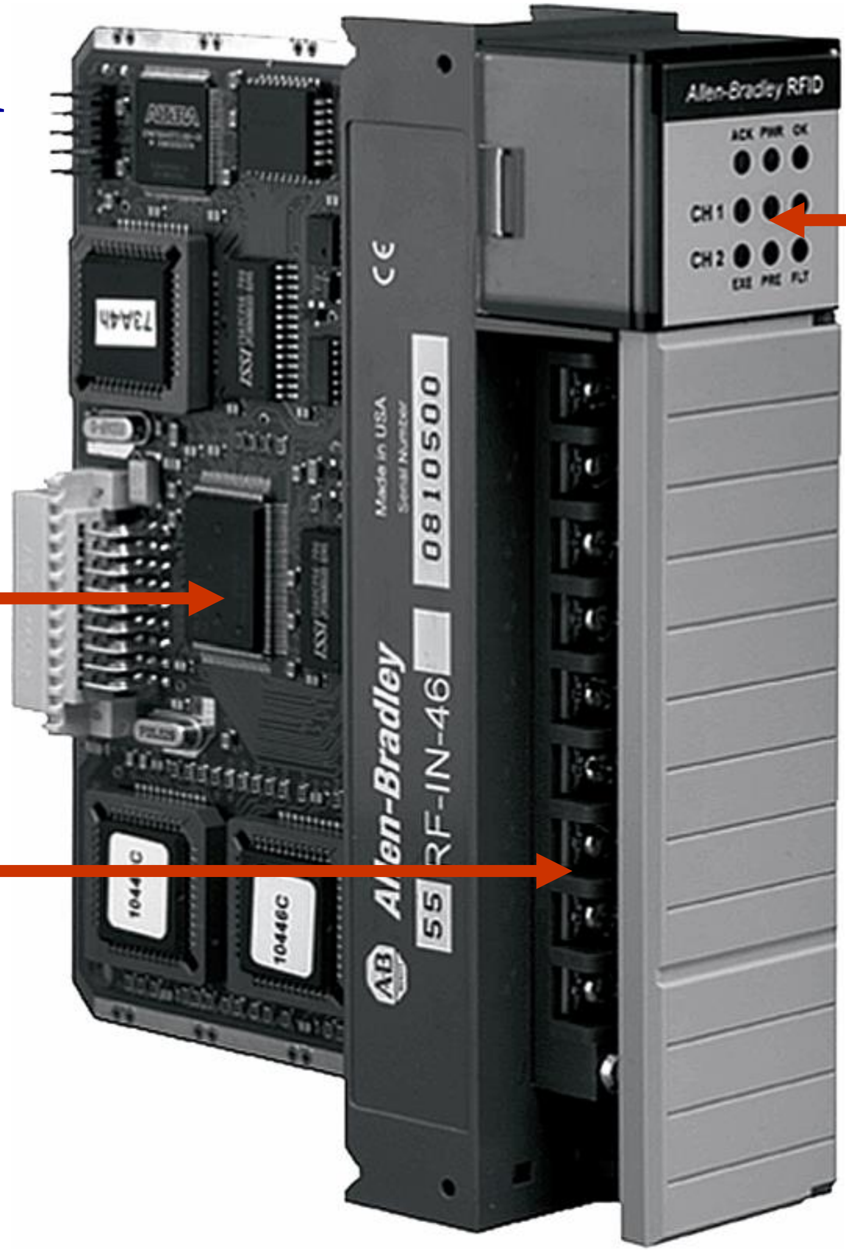


This implementation uses an *input/output card* in conjunction with the PC as an interface for the field devices.

Combination I/O modules **can have both input and output connections in the same physical module.**



An I/O module is made up of a printed circuit board and a terminal assembly.



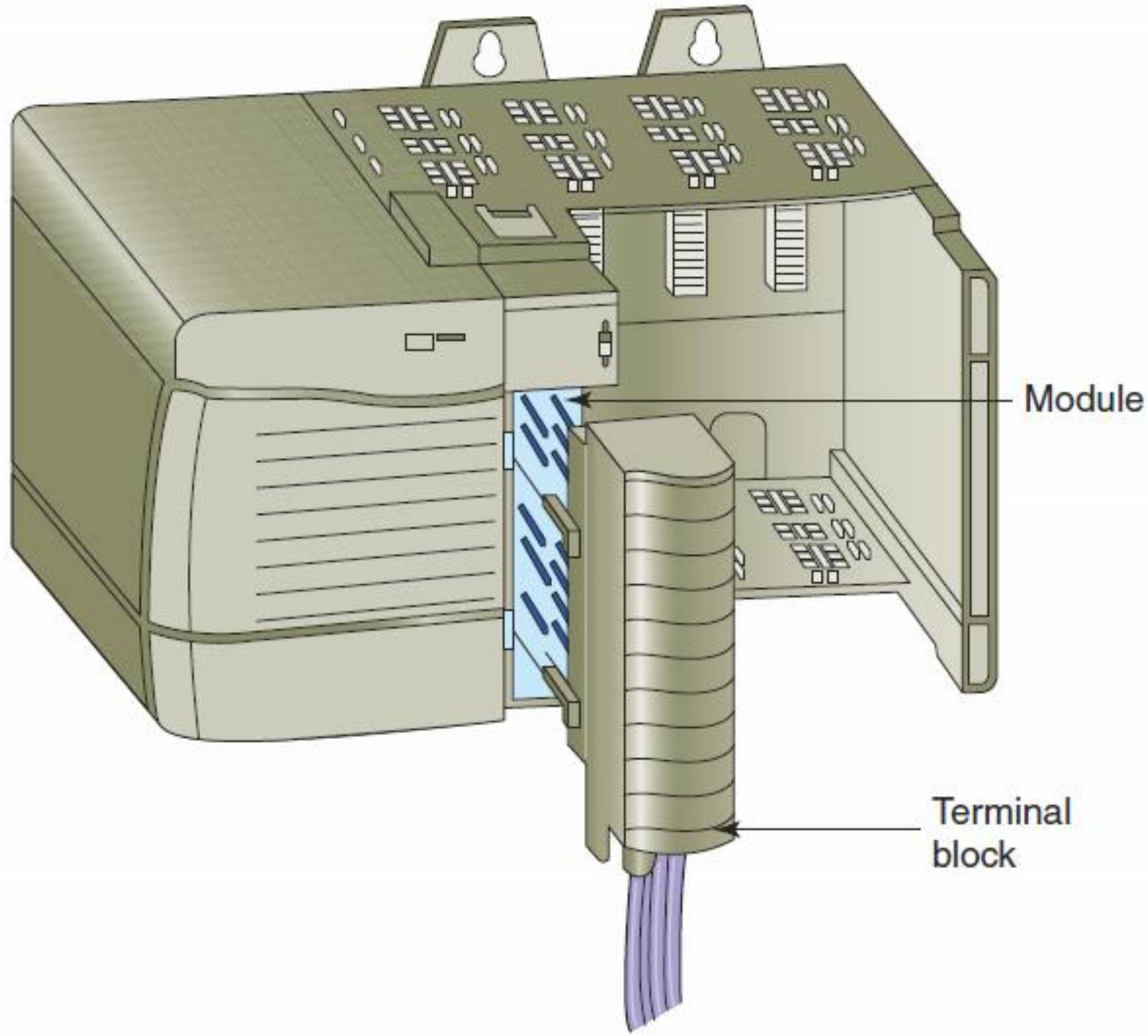
Status Lights

The printed circuit board contains electronic circuitry.

The terminal assembly is used for making field-wiring connections.

Most PLC modules have *plug-in wiring terminal strips* that plug into the actual module.

If there is a problem with a module, the entire strip is removed, a new module is inserted, and the terminal block is plugged into the new module.



I/O modules can be 8, 16, 32, or 64 point cards.

- **The *points* refers to the number of inputs or outputs available.**

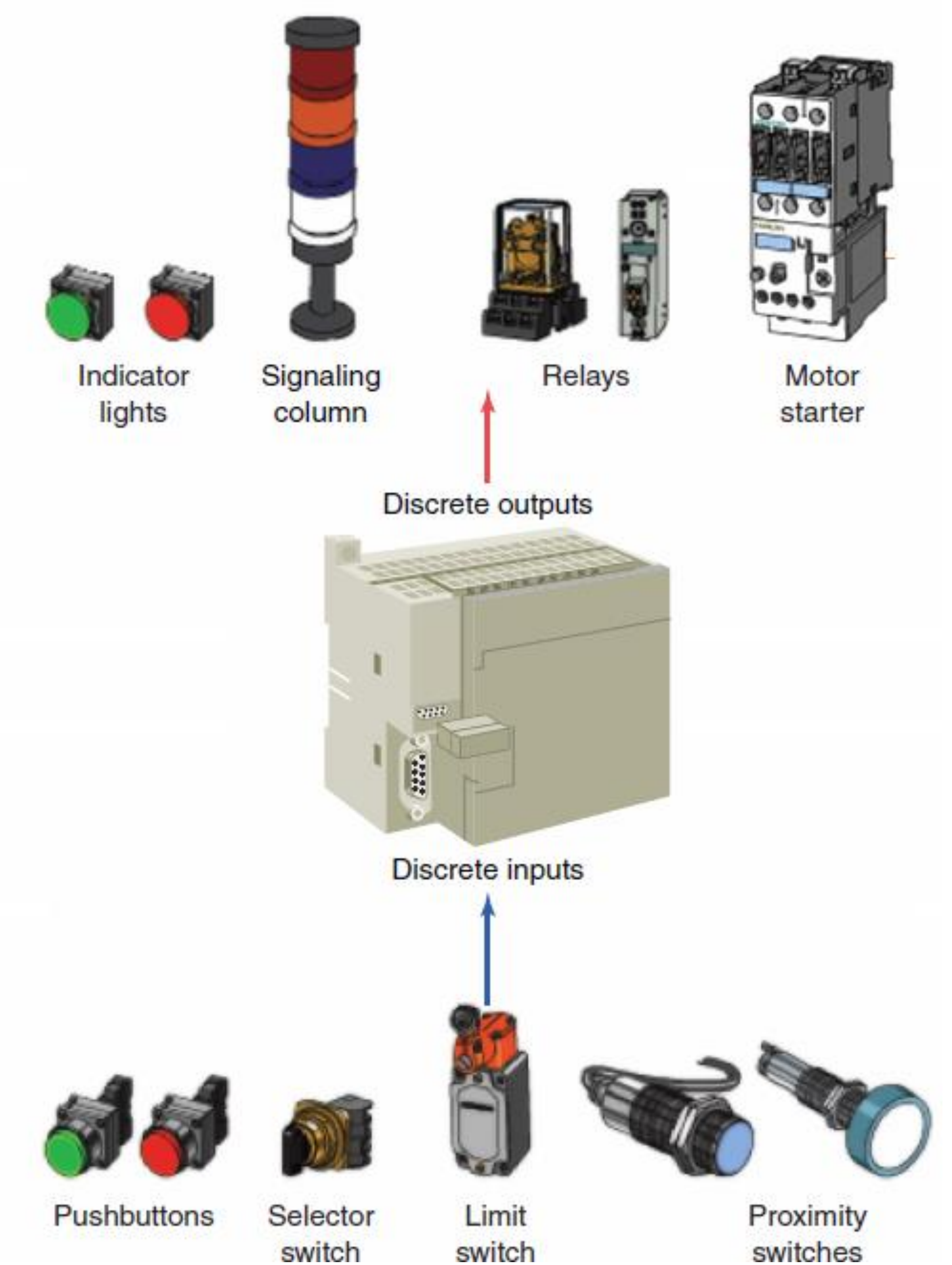
  - **A *high-density* 64 point card provides a greater space saving but with less rated current output per output.**

2.2

Discrete I/O Modules

The *discrete I/O* interface module connects field input devices of the *ON/OFF* nature.

The classification of discrete I/O covers *bit oriented* inputs and outputs.



Each discrete I/O module is powered by some *field supplied* voltage source.

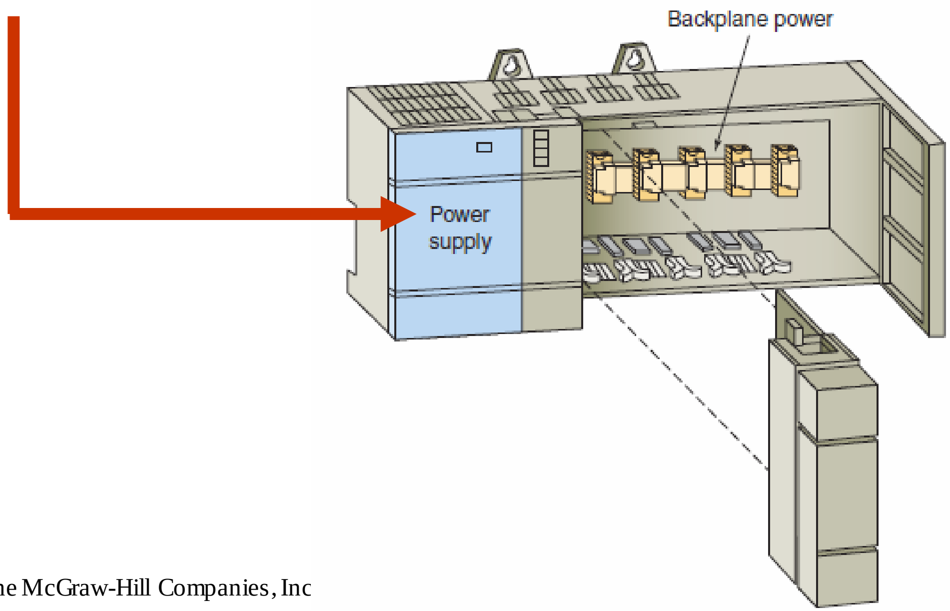
Since these voltages can be of different magnitude or type, I/O modules are available at various AC and DC voltage ratings.

Table 2-1 Common Ratings for Discrete I/O Interface Modules

Input Interfaces	Output Interfaces
12 V **AC/DC** /24 V **AC/DC**	12–48 V **AC**
48 V **AC/DC**	120 V **AC**
120 V **AC/DC**	230 V **AC**
230 V **AC/DC**	120 V **DC**
5 V **DC** (TTL level)	230 V **DC**
	5 V **DC** (TTL level)
	24 V **DC**

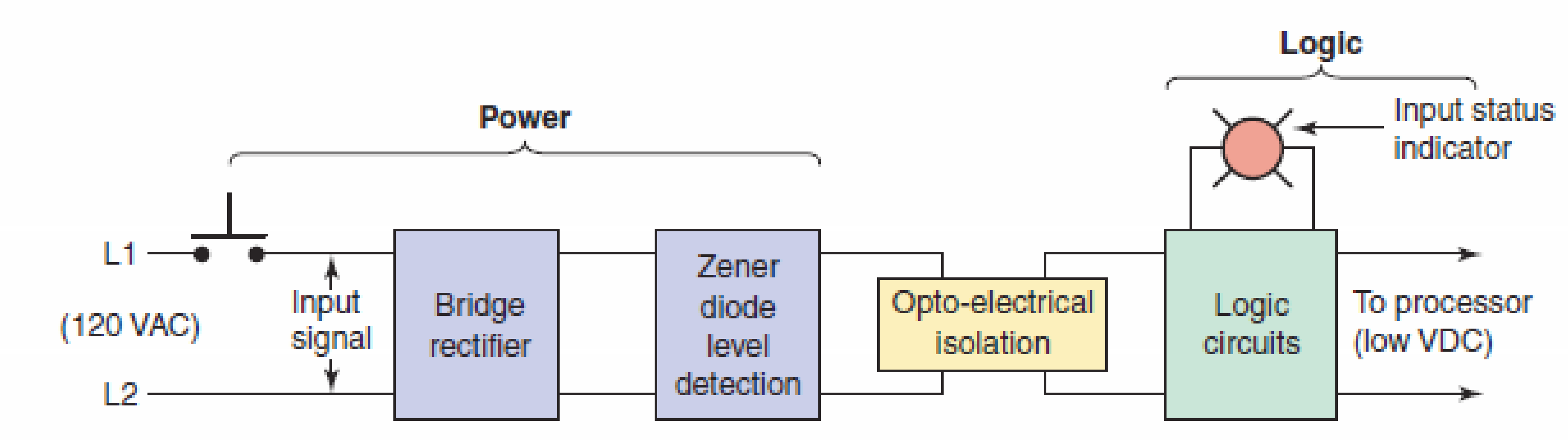
The modules themselves receive their voltage and current for proper operation from the *backplane* of the rack enclosure into which they are inserted.

Backplane power is provided by the PLC module power supply and is used to power the electronics that reside on the I/O module circuit board.



Currents required by the loads are normally provided by user-supplied power.

AC *discrete input module* block diagram.

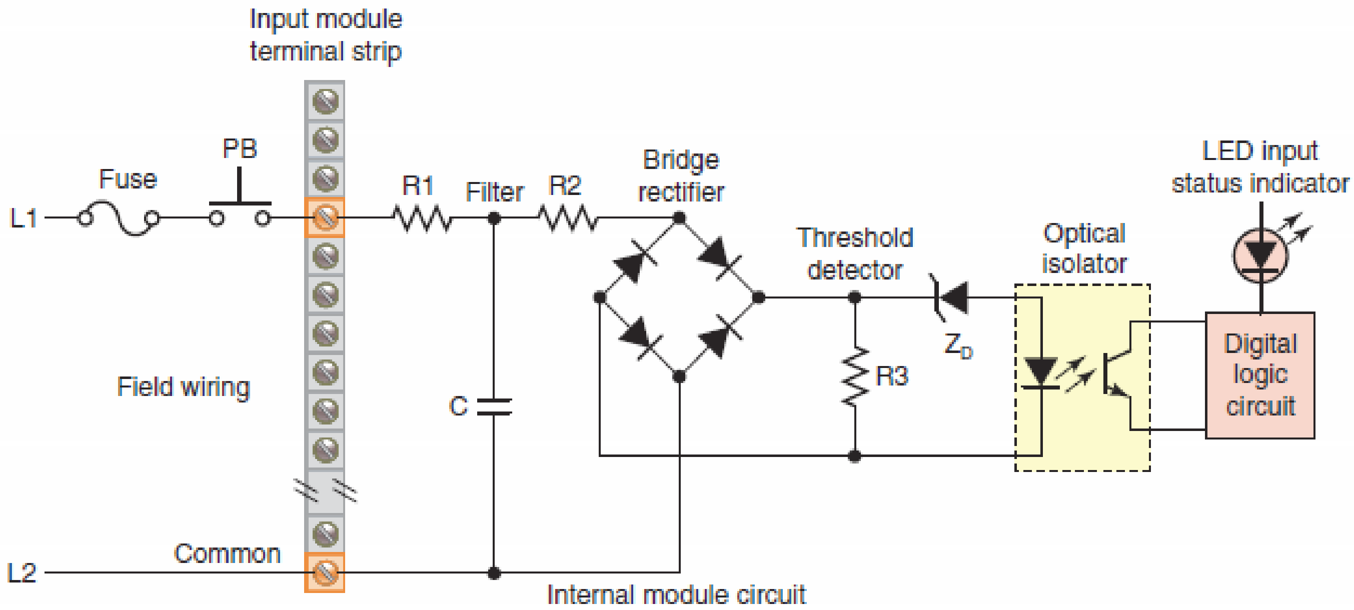


➢The circuit is made up of a *power* section and a *logic* section.

➢An *optical isolator* is used to provide electrical isolation between the power and logic circuitry.

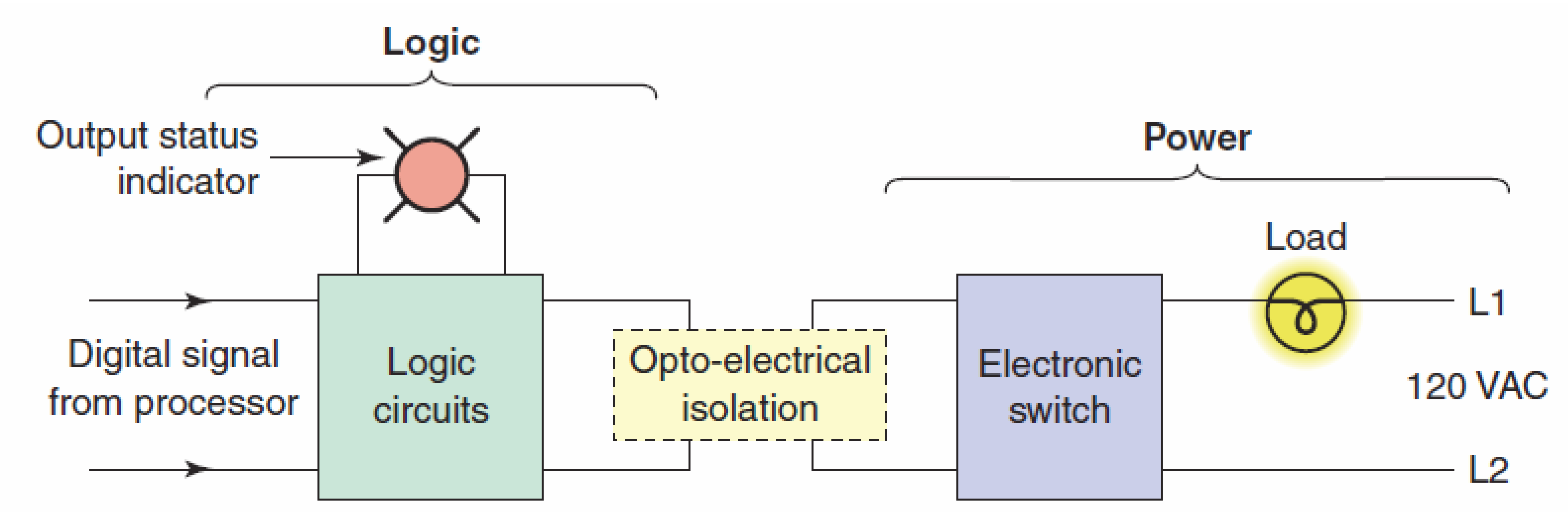
➢The input LED turns on or off, indicating the *status* of the input device.

AC *discrete input module* schematic diagram.



- ➢When the pushbutton is closed AC is applied to the bridge rectifier input.
- ➢This results in a low-level DC output voltage that is applied across the LED of the optical isolator.
- ➢When light from the LED strikes the phototransistor, it switches it into conduction.

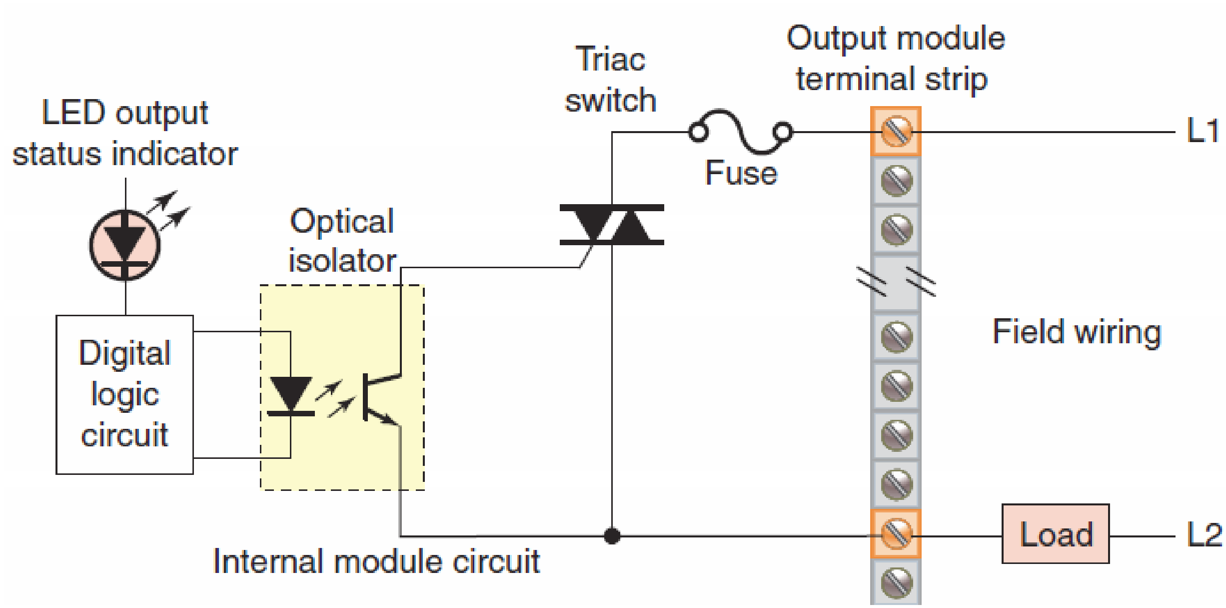
AC *discrete output module* block diagram.



➢**The module is composed of the power section and the logic section, coupled by an isolation circuit.**

➢**The power output interface can be thought of as an electronic switch that turns the output load device on and off.**

AC *discrete output module* schematic diagram.

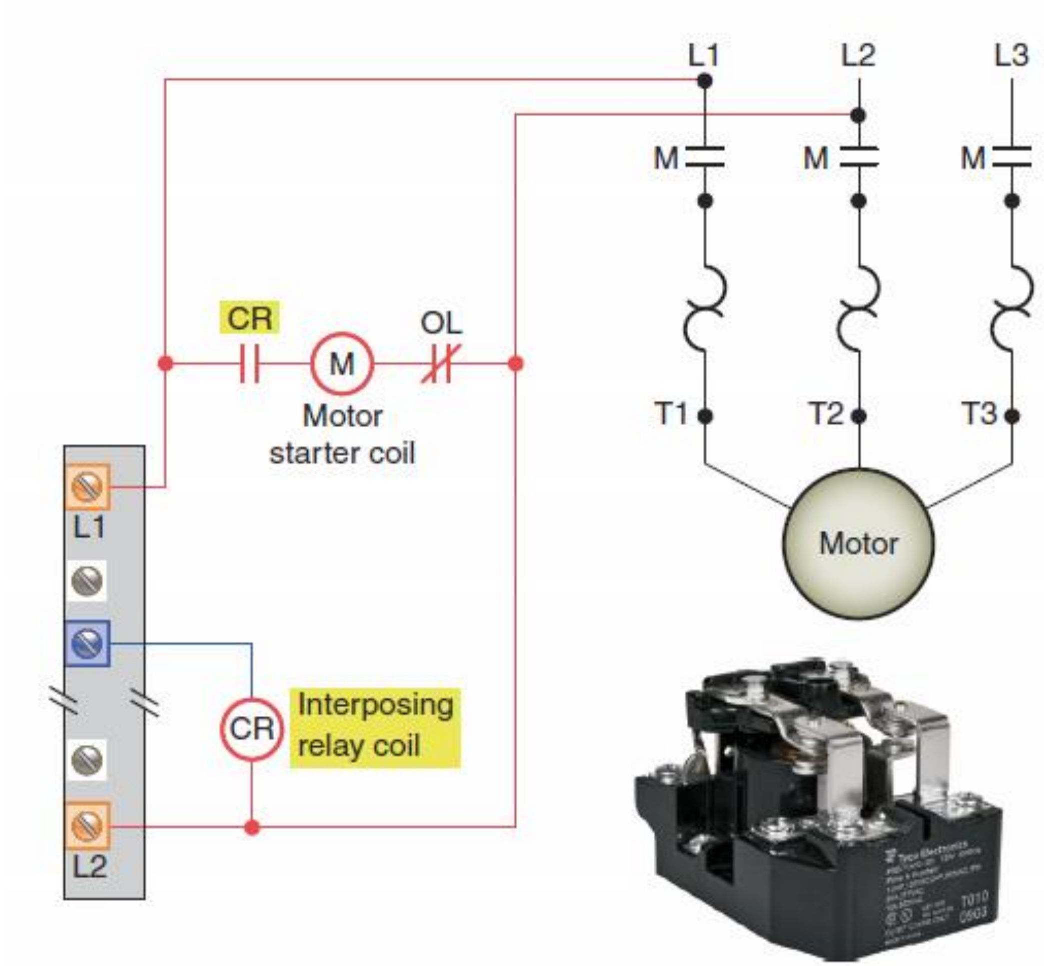


➢**When the processor calls for the output load to be energized, a voltage is applied across the LED of the opto-isolator.**

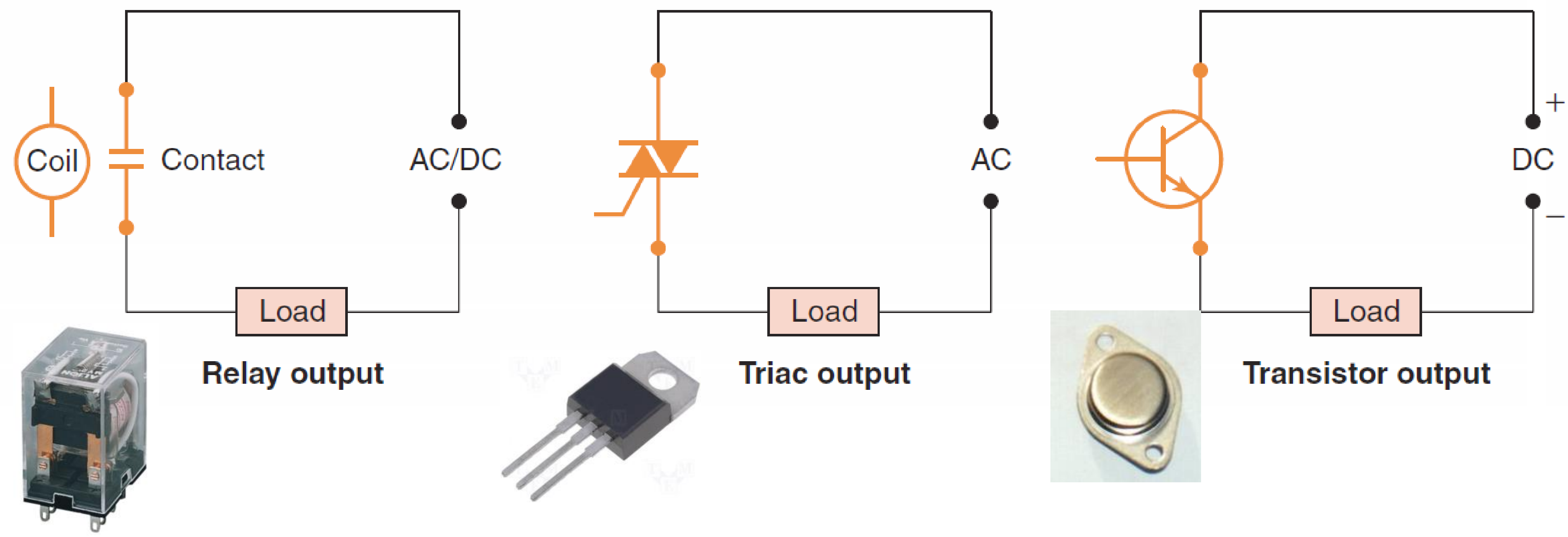
➢**This in turn triggers the triac AC semiconductor switch into conduction allowing current to flow to the output load.**

Individual AC outputs are usually limited by the size of the triac to *1 A or 2 A*.

For controlling larger loads an ***interposing relay*** is connected to the output module. The contacts of the relay can then be used to control a larger load or motor starter,

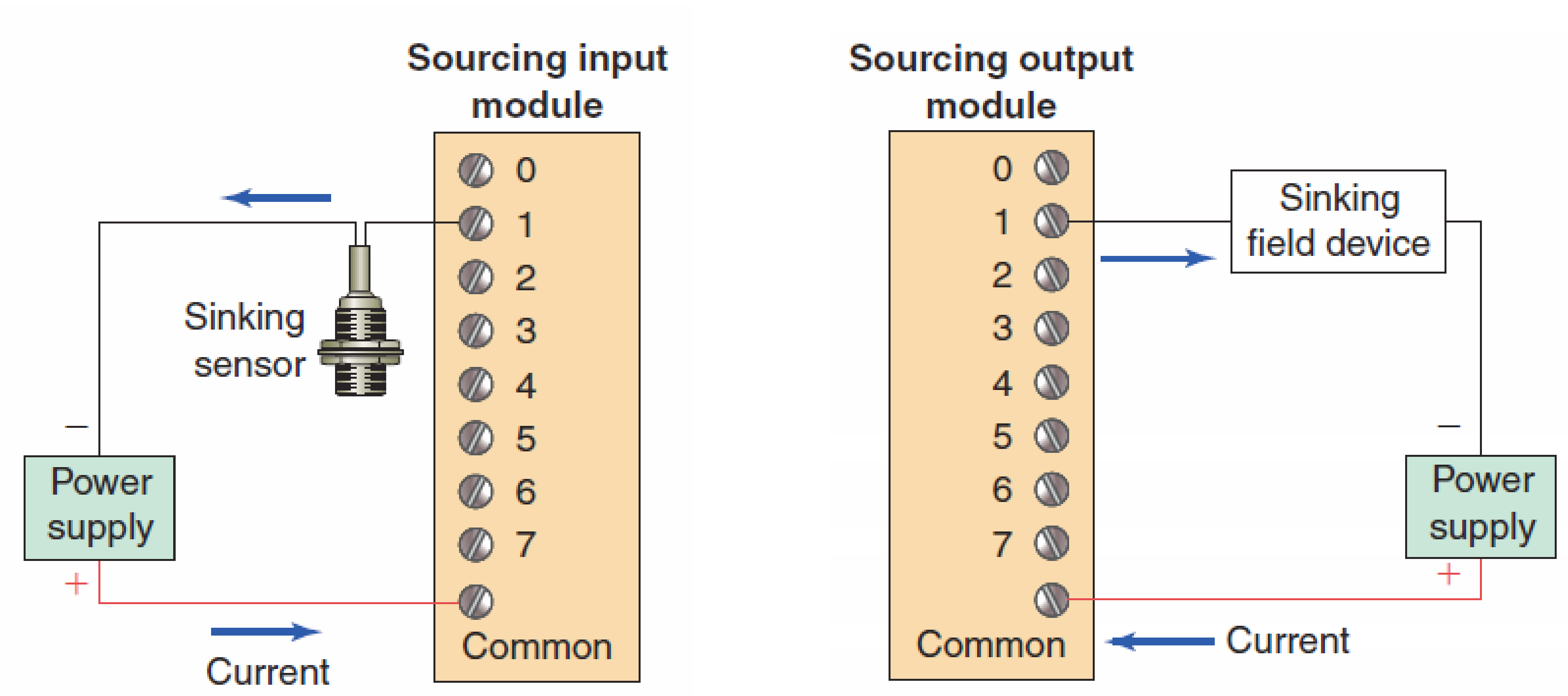


Discrete output modules are used to turn field output devices either *on or off*.



Output modules can be purchased with *transistor*, *triac*, or *relay* output

Certain DC I/O modules specify whether the module is designed for interfacing with *current-source* or *current-sink* devices.



If the module is a *current-sourcing module*, then the input or output device must be a *current-sinking device*.

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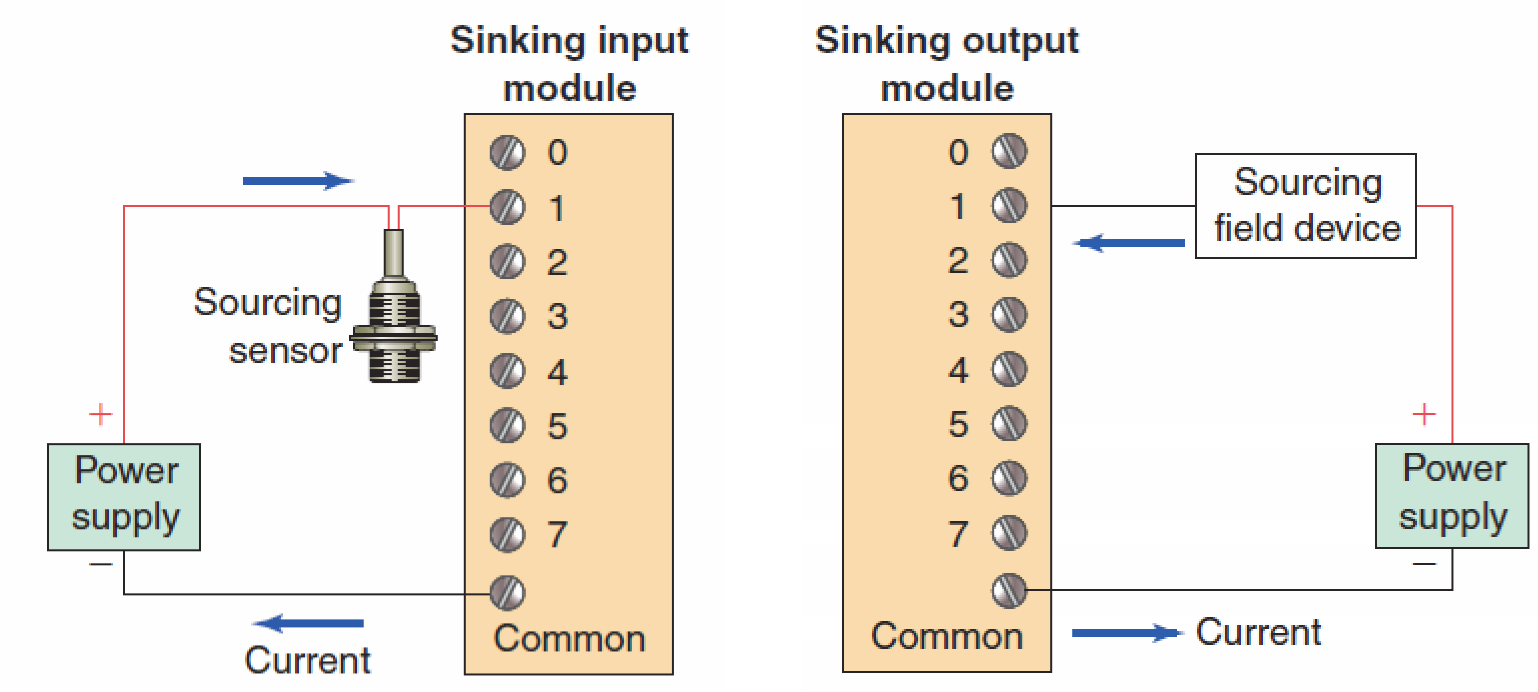
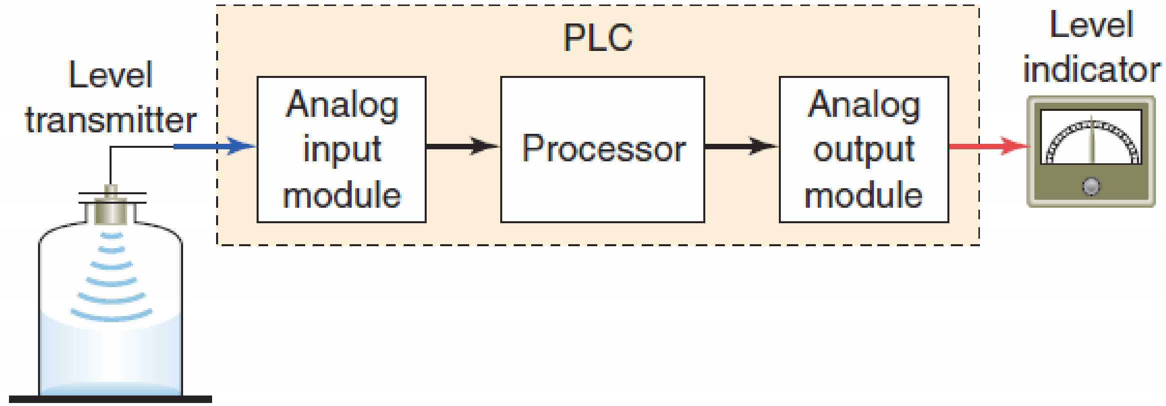


2.3

Analog I/O Modules

***Analog* input and output devices have an infinite number of values.**

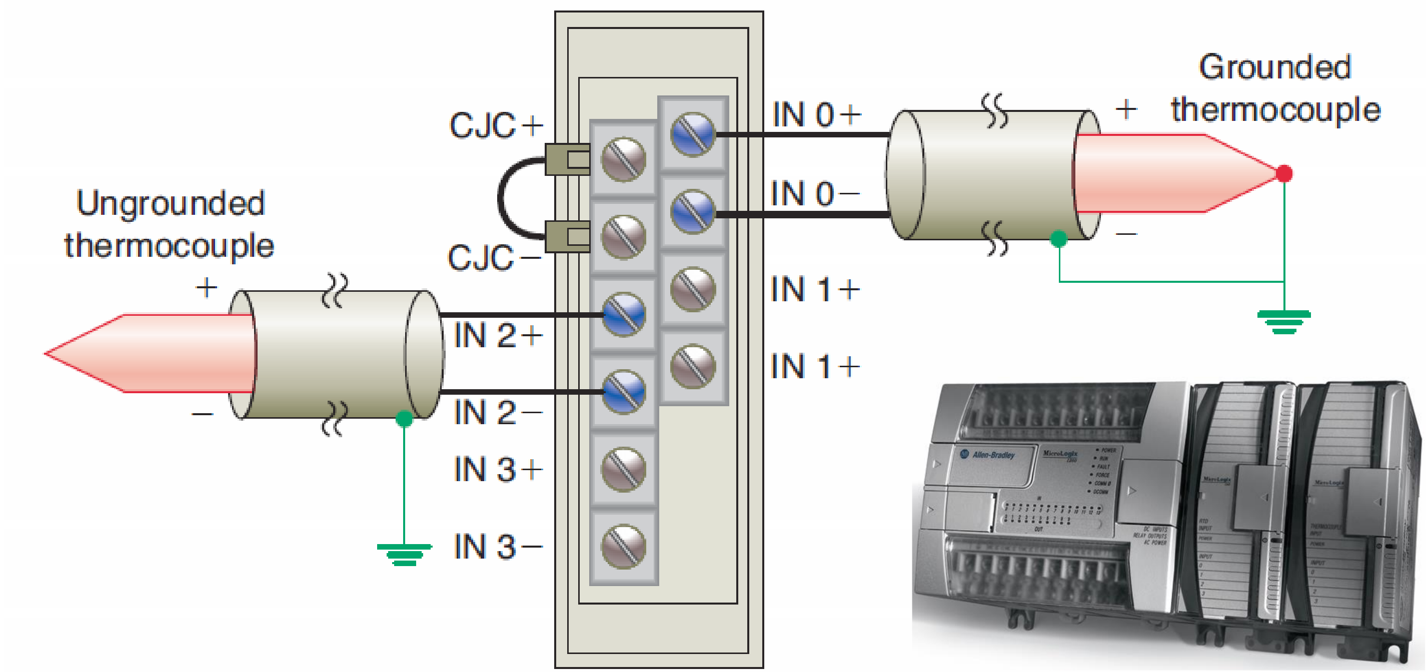


- **Typical analog inputs and outputs vary from *0 to 20 milliamps*, *4 to 20 milliamps*, or *0 to 10* volts.**

- **The analog input interface module accepts an analog signal and converts it to a digital signal.**

- **The analog output module accepts a digital signal and converts it to an analog signal that operates the output.**

The two basic types of *analog input modules* are *voltage* sensing and *current* sensing.



A varying DC voltage in the low millivolt range, proportional to the temperatures being monitored, is produced by the thermocouples.

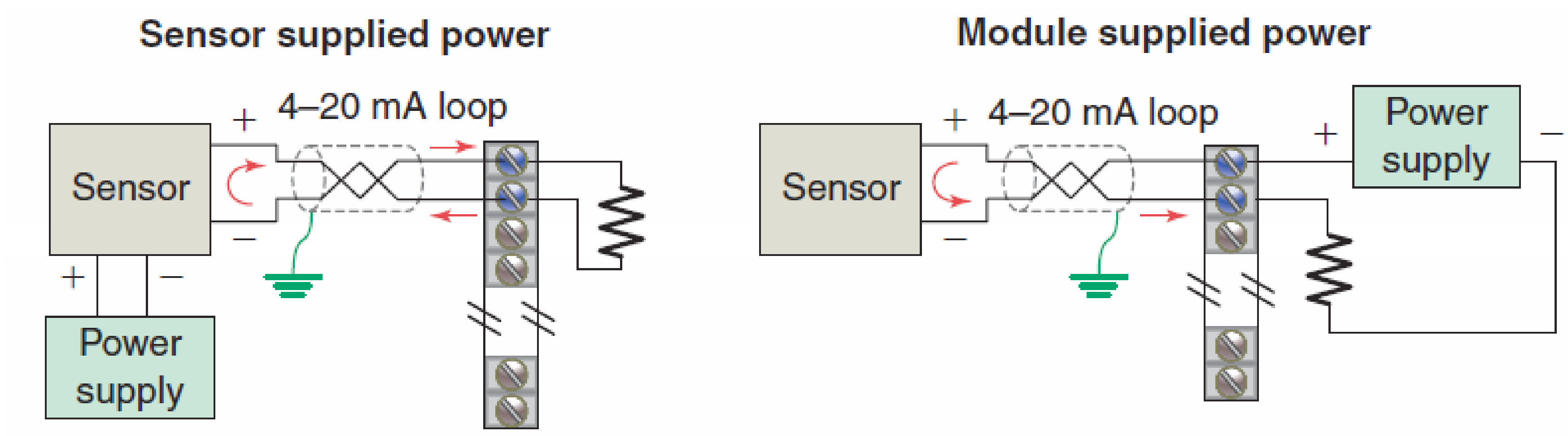
The main element of the analog input module is an *analog-to-digital* (A/D) converter

- ***Bipolar*** **input modules can accept signals that swing between a negative and positive value.**
- ***Unipolar*** **input modules can accept an input signal that varies in the positive direction only.**
- ***Resolution*** **refers to the smallest change in input signal value that can be sensed**

Span of analog input	Bipolar	10 V	−10 to +10 V
		5 V	−5 to +5 V
	Unipolar	10 V	0 to +10 V
		5 V	0 to +5 V
Resolution			0.3 mV

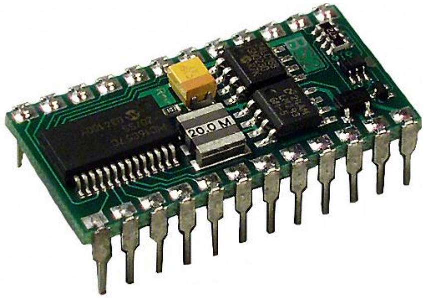
Unlike voltage input signals, current signals are not as sensitive to noise and typically are not distance limited.

The current sensing loop power may be supplied by the sensor or the analog output module.



Shielded twisted pair cable is normally recommended for connecting any type analog input signal.

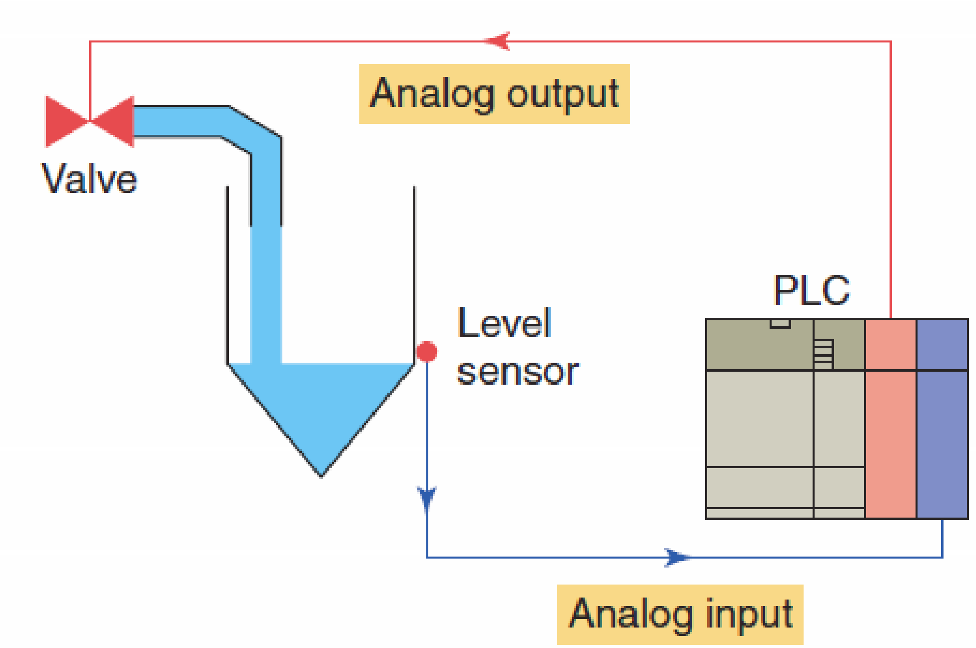
The main element of the analog output module is an *digital-to-analog* (D/A) converter



The *analog output module* receives from the processor digital data, which are converted into a proportional voltage or current to control an analog field device.

The analog output signal is varied under the *control of the PLC program* and can be used for control of an analog control valve.

Analog I/O control system



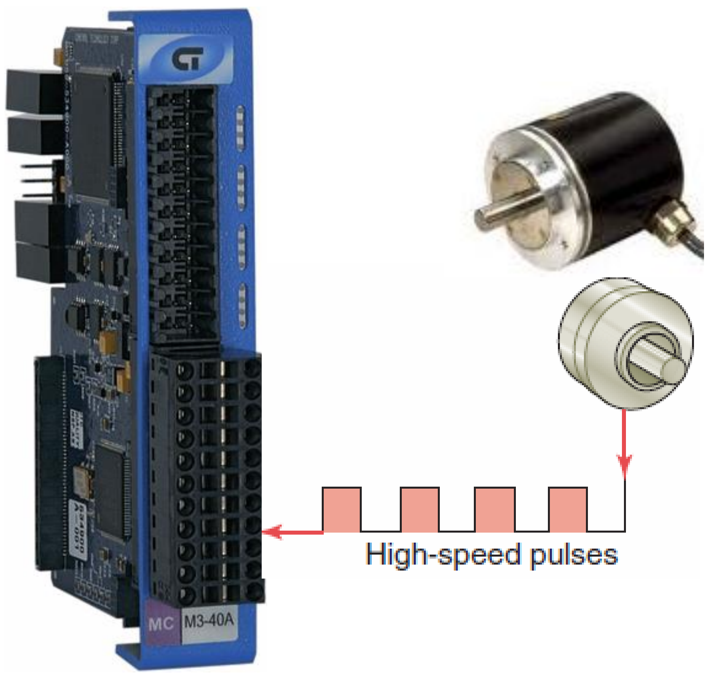
The PLC controls the amount of fluid placed in a holding tank by adjusting the percentage of the valve opening.

- The valve is initially opened 100 percent.
- As the fluid level in the tank approaches the preset point, the processor modifies the output, which adjusts the valve to maintain a set point.

2.4

Special I/O Modules

High-speed counter modules **are used to count pulses from sensors, encoders, and switches that operate at very high speeds.**



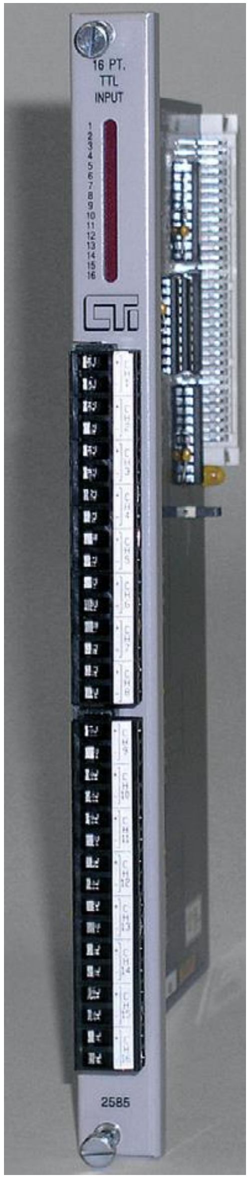
They have the electronics needed to count independently of the processor.

A typical count rate available is 0 to 100 kHz, which means the module would be able to count *100,000 pulses per second.*

The *thumbwheel module* allows the use of thumbwheel switches for feeding information to the PLC to be used in the control program.

The *TTL module* allows the transmitting and receiving of TTL (Transistor-Transistor-Logic) signals.

This module allows devices that produce TTL-level signals to communicate with the PLC's processor

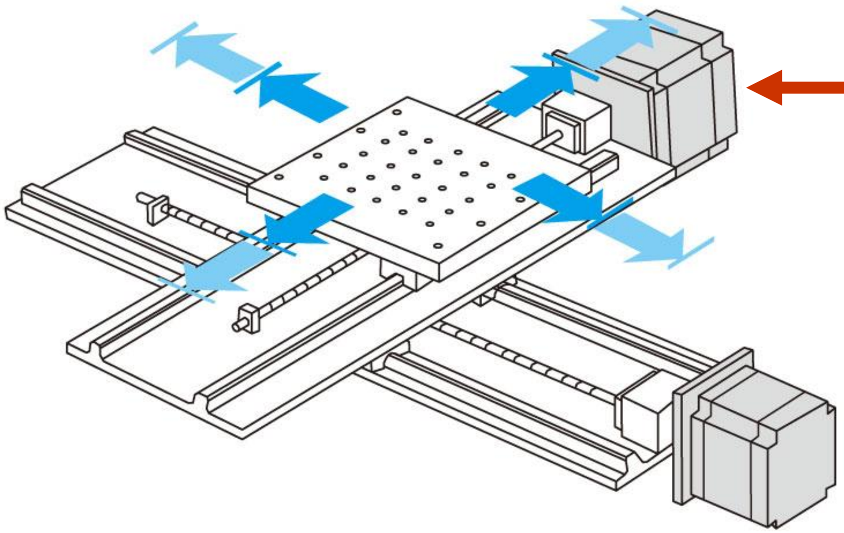


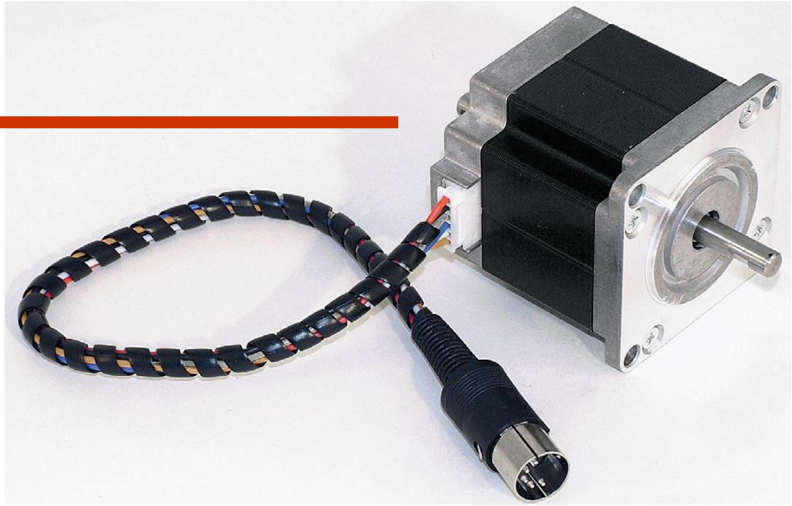
An *encoder-counter module* allows the user to read the signal from an encoder on a real-time basis and stores this information so it can be read later by the processor.

The *BASIC or ASCII module* runs user written BASIC and C programs.

Typical applications include interfaces to bar code readers, robots, printers, and displays.

The *stepper-motor module* provides pulse trains to a stepper-motor translator, which enables control of a stepper motor.

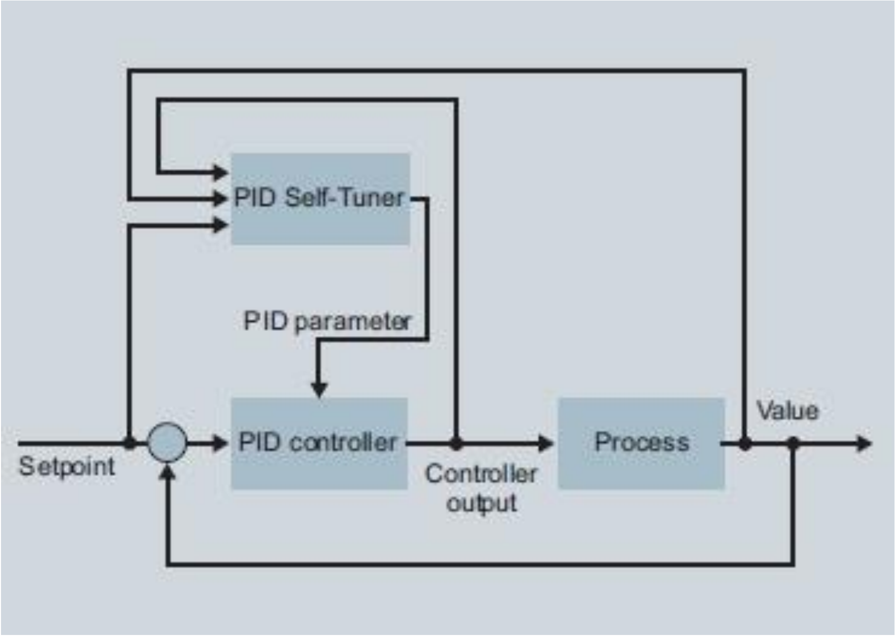




The commands for the module are determined by the control program in the PLC.

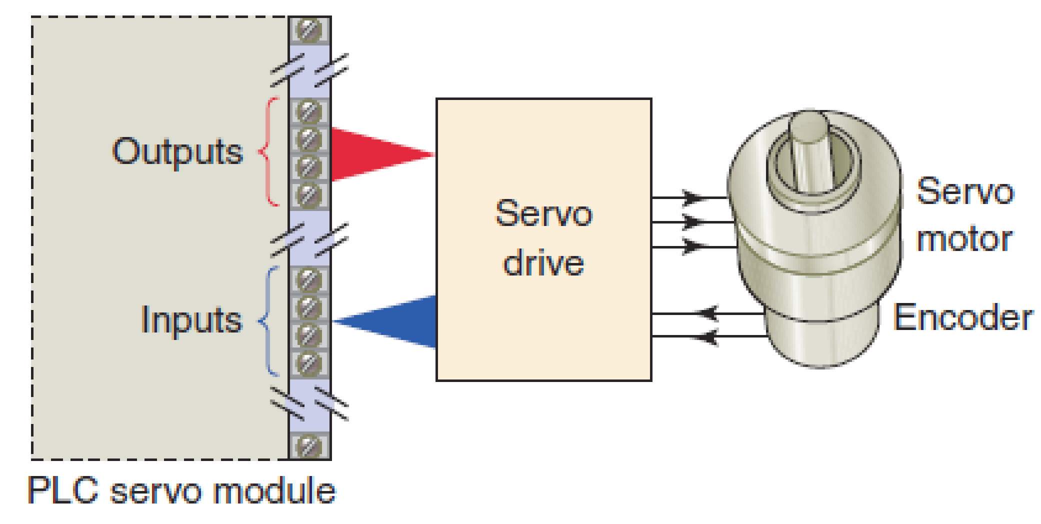
The *BCD-output module* enables a PLC to operate devices that require BCD-coded signals such as seven-segment displays

The *proportional-integral-derivative (PID) module* is used in process control applications that incorporate PID algorithms.



This arrangement prevents the CPU from being burdened with complex calculations.

Motion and position control modules **are used in applications involving accurate high-speed machining and packaging operations.**



Intelligent position and motion control modules permit PLCs to control stepper and servo motors.

***Serial communications modules* are used to establish point-to-point connections with other intelligent devices for the exchange of data.**

Communication modules provide for connection to PLC networks.

2.5

I/O Specifications

Discrete I/O Module Specifications

Nominal Input Voltage - Specifies the magnitude (e.g., 5 V, 24 V, 230 V) and type (AC or DC) of user-supplied voltage that a module is designed to accept.

Input Threshold Voltages - Specifies the minimum ON-state voltage at which logic 1 is recognized and the maximum OFF-state voltage at which logic 0 is recognized.

Nominal Current Per Input - Specifies the minimum input current that the discrete input devices must be capable of driving to operate the input circuit.

Discrete I/O Module Specifications

Ambient Temperature Rating - Specifies what the maximum temperature of the air surrounding the I/O modules should be for best operating conditions.

Input ON/OFF Delay (response time) - Specifies the maximum time duration required by an input module's circuitry to recognize that a field device has switched ON (input ON-delay) or switched OFF (input OFF-delay).

Output Voltage - This AC or DC value specifies the magnitude (e.g., 5 V, 115 V, 230 V) and type (AC or DC) of user-supplied voltage at which a discrete output module is designed to operate.

Discrete I/O Module Specifications

Output Current - Specifies the maximum current that a single output and the module as a whole can safely carry under load (at rated voltage).

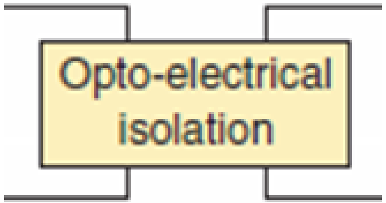
Inrush Current - Specifies the maximum inrush current and duration (e.g., 20 A for 0.1 s) for which an output circuit can exceed its maximum continuous current rating.

Short Circuit Protection - Specifies whether the particular output module's design has individual protection for each circuit or if fuse protection is provided for groups (e.g., 4 or 8) of outputs.

Discrete I/O Module Specifications

Leakage Current - This value specifies the amount of current still conducting through an output circuit even after the output has been turned off.

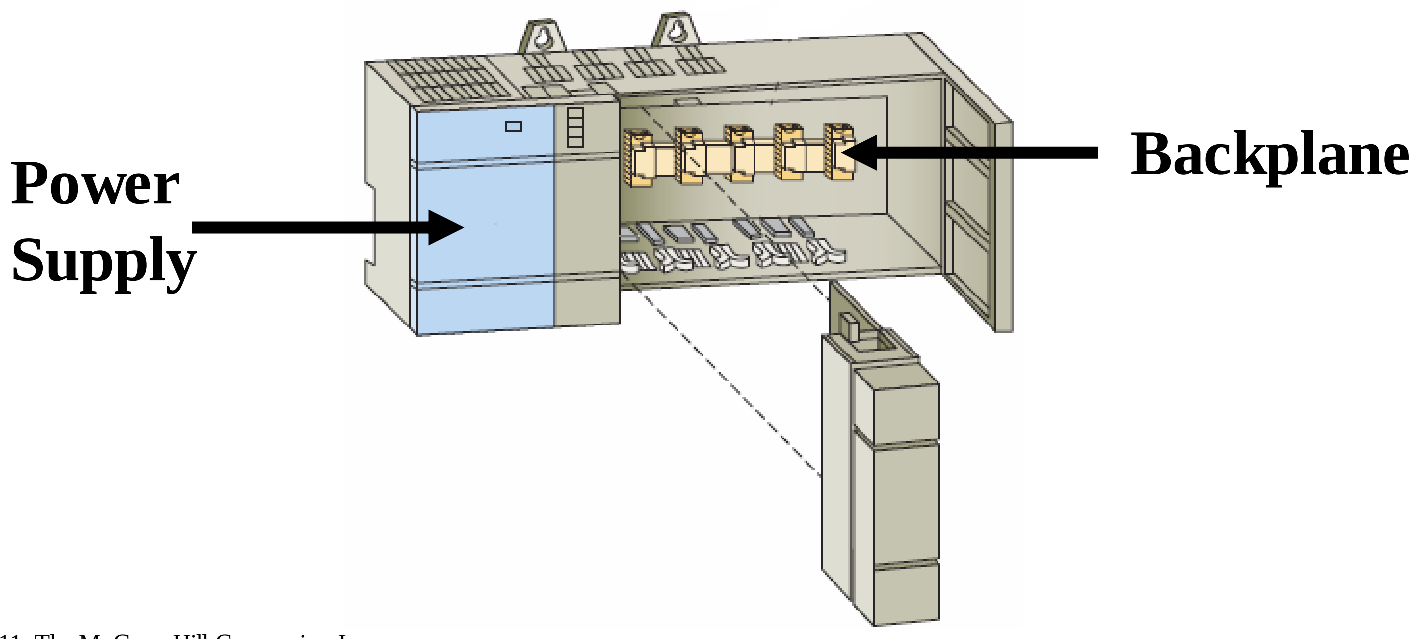
Electrical Isolation - The specification for electrical isolation, typically 1500 or 2500 volts AC, rates the module's capacity for sustaining an excessive voltage at its input or output terminals.



Points Per Module - This specification defines the number of field inputs or outputs that can be connected to a single module.

Discrete I/O Module Specifications

Backplane Current Draw - **This value indicates the amount of current the module requires from the backplane. The sum of the backplane current drawn for all modules in a chassis is used to select the appropriate chassis power supply rating.**



Analog I/O Module Specifications

Channels Per Module - **Whereas individual circuits on discrete I/O modules are specified as points per module, circuits on analog I/O modules are specified as channels per module.**

Input Current/Voltage Range(s) - **These are the voltage or current signal ranges that an analog input module is designed to accept.**

Output Current/Voltage Range(s) - **This specification defines the current or voltage signal ranges that a particular analog output module is designed to output under program control.**

Analog I/O Module Specifications

Input Protection - **Analog input circuits are usually protected against accidentally connecting a voltage that exceeds the specified input voltage range.**

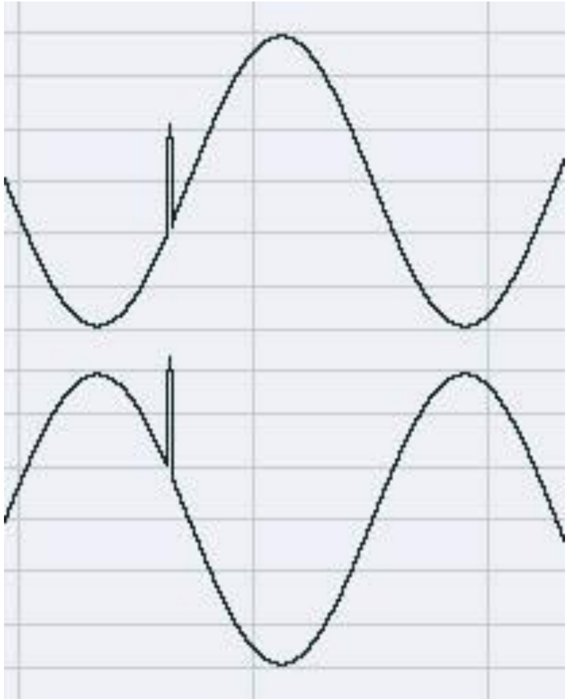
Resolution - **The resolution specifies how accurately an analog value can be represented digitally. This will determine the smallest measurable unit of current or voltage change that can be detected.**

Input Impedance and Capacitance - **For analog I/Os, these values must be matched to the external device connected to the module. Typical ratings are in the Megohm and Picofarad range.**

Analog I/O Module Specifications

Common-Mode Rejection - Noise is generally caused by electromagnetic interference, radio frequency interference, and ground loops.

Noise that is picked up equally in parallel wires is rejected because the difference is zero. Twisted pair wires are used to ensure that this type of noise is equal on both wires.



2.6

The Central Processing Unit (CPU)

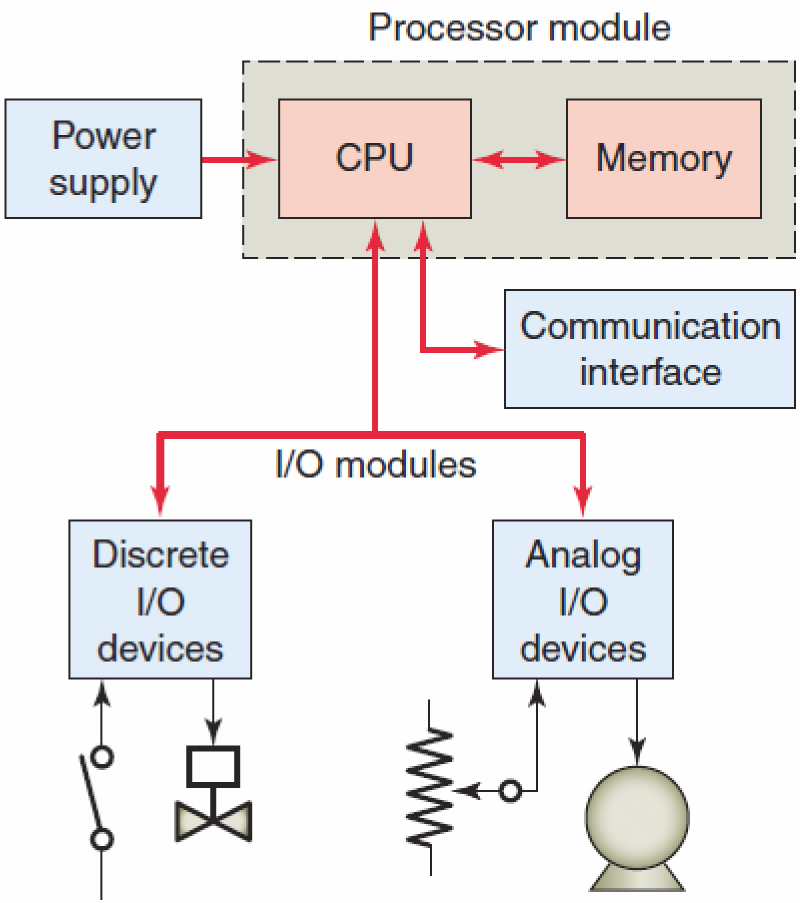
The *central processing unit (CPU)* is built into fixed PLCs while modular types typically use a plug-in module.

***CPU*, *controller*, and *processor* are all terms used by different manufacturers to denote the same module that performs basically the same functions.**

A processor module can be divided into two sections: the *CPU* section and the *memory* section

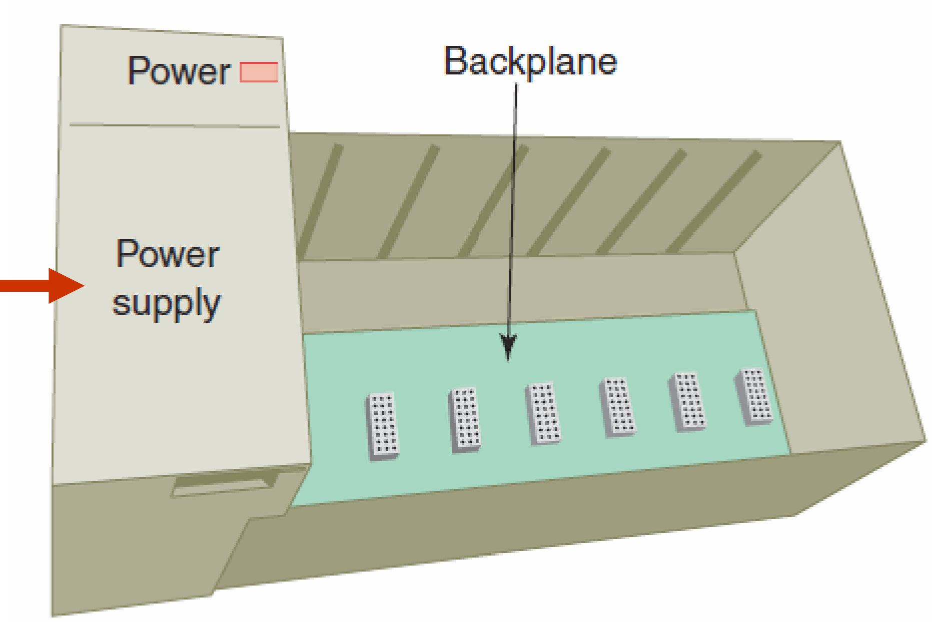
The **CPU** executes the program.

The **memory** stores the program along with other retrievable data.

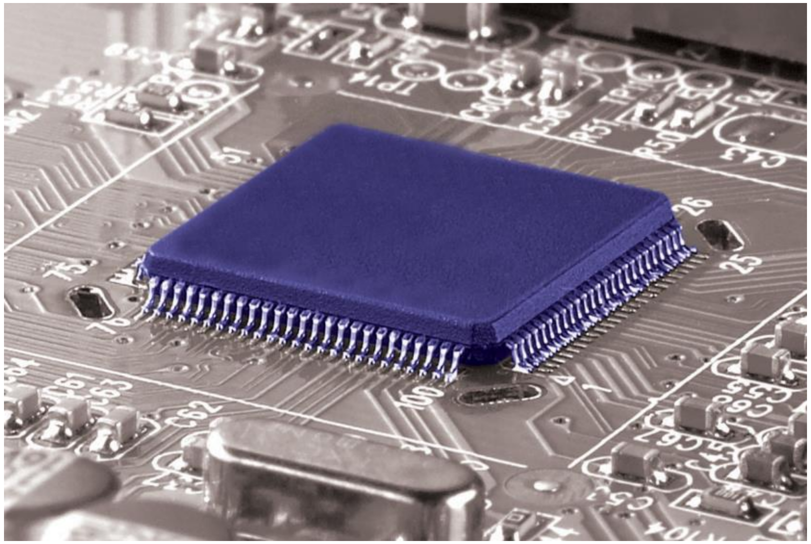


The *PLC power supply* provides the necessary power to the processor and I/O modules plugged into the backplane of the rack.

The power supply converts the AC input voltage into the usable DC voltage required by the CPU, memory, and I/O electronic circuitry.

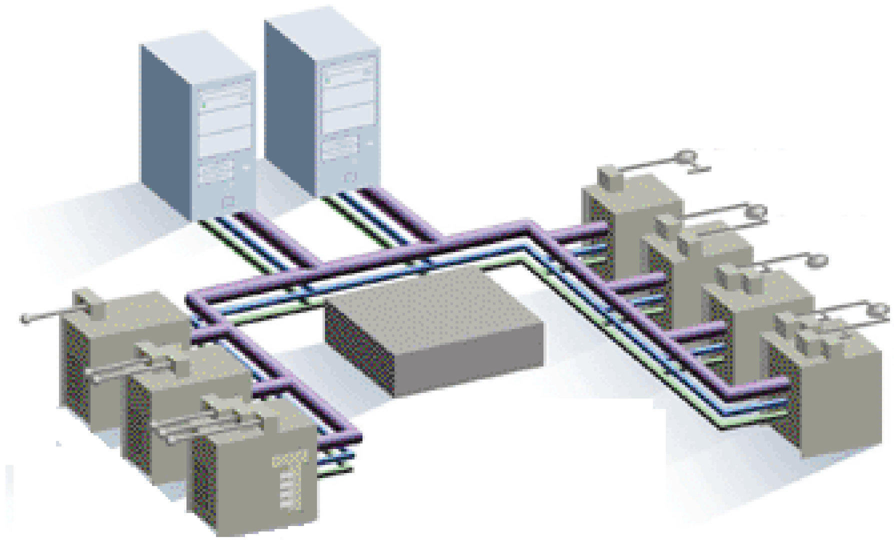


The CPU contains the similar type of *microprocessor* found in a personal computer.



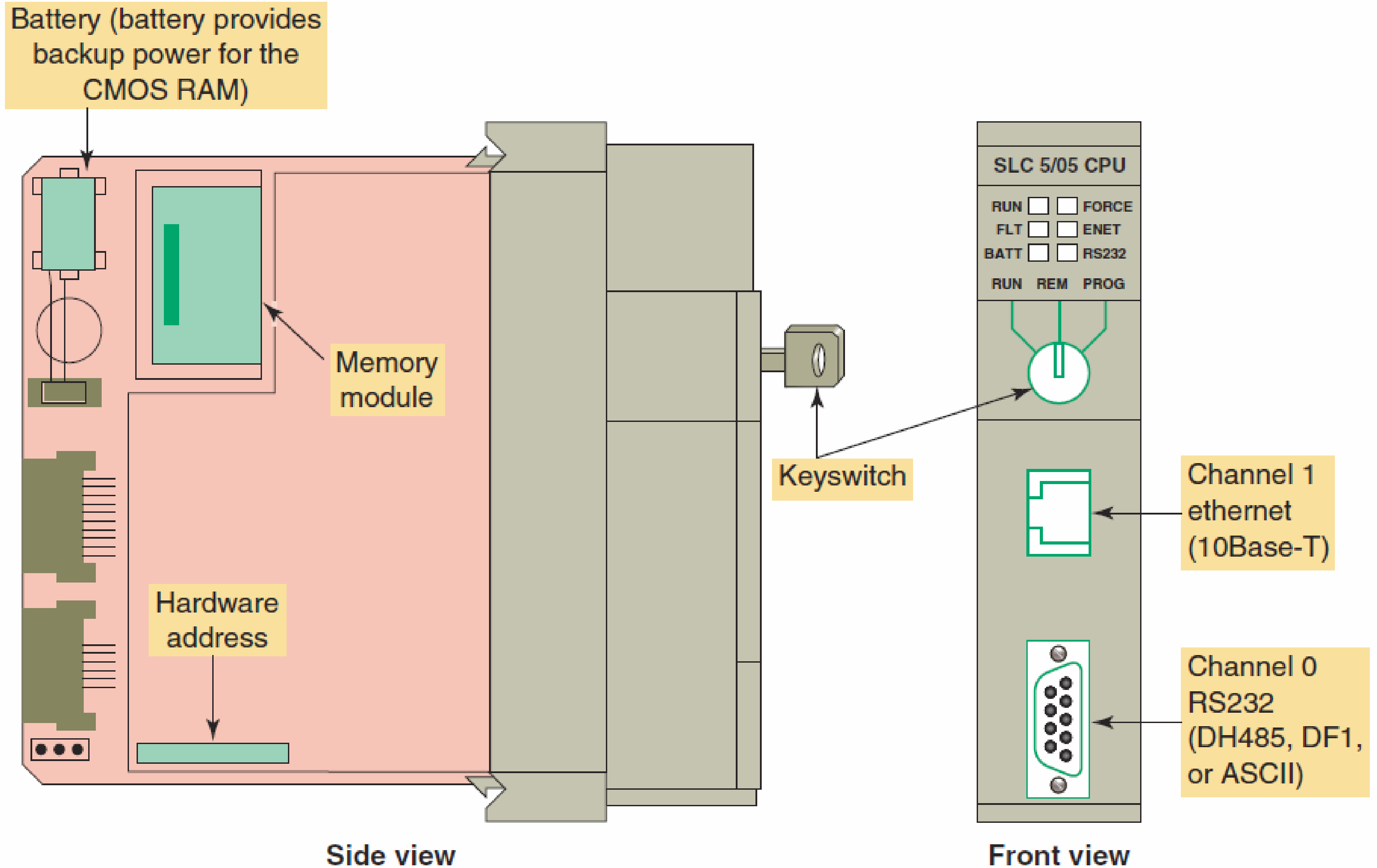
A PLC microprocessor is designed to facilitate industrial control rather than provide general purpose computing.

The CPU of a PLC system may contain *more than one* processor.



Fault-tolerant PLC systems support dual processors for critical processes. These systems allow the user to configure the system with *redundant* (two) processors, which allows transfer of control to the second processor in the event of a processor fault.

Associated with the processor unit will be a number of status LED indicators to provide system diagnostic information to the operator.



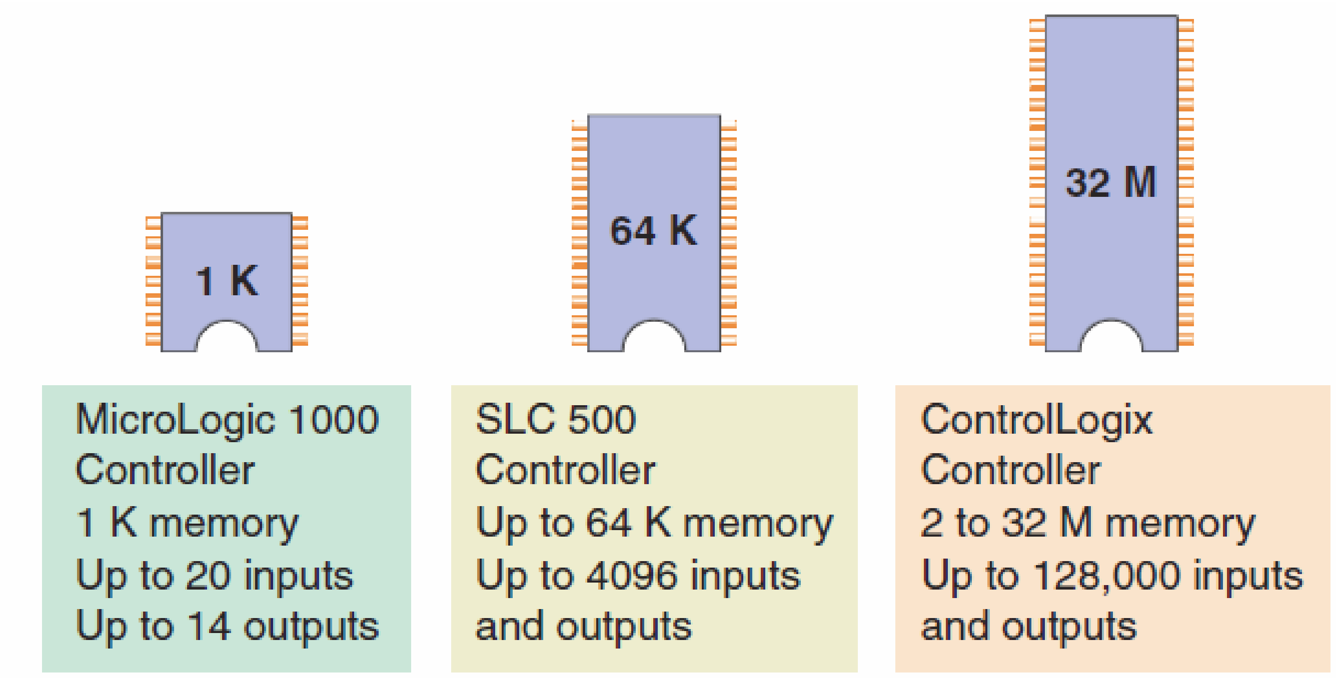
Many electronic components found in processors and other types of PLC modules are sensitive to *electrostatic* voltages that can degrade their performance or damage them.

When not in use, store modules in a static-shield bag.

2.7

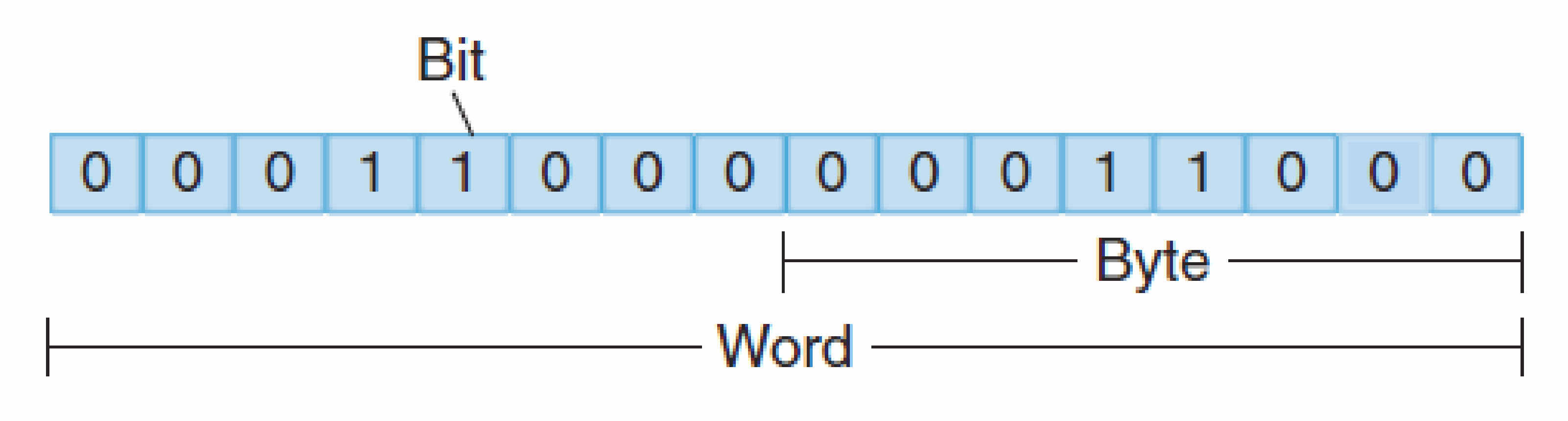
Memory Design

Memory **is the element that stores information, programs, and data in a PLC.**



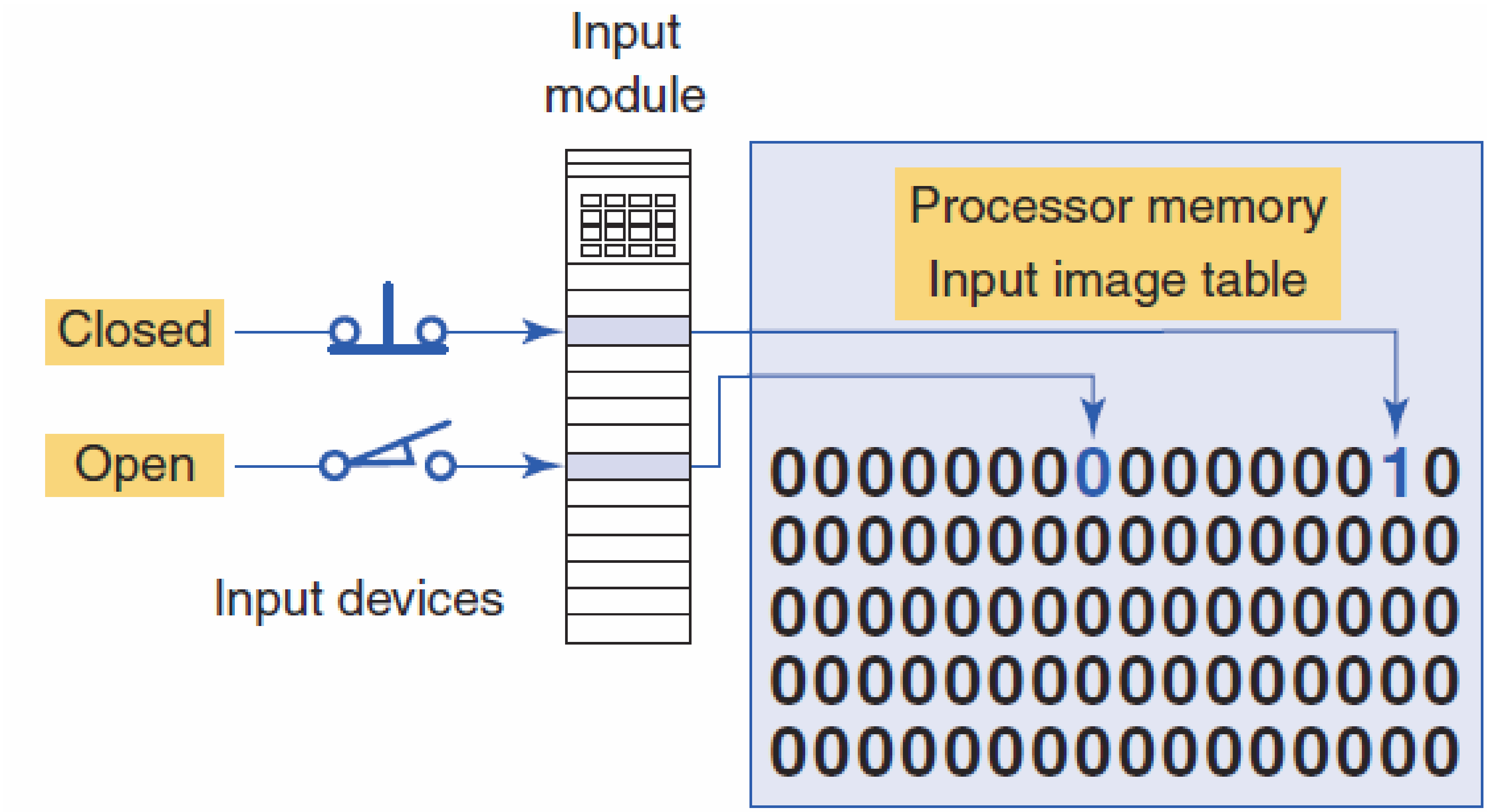
The *complexity* of the program determines the amount of memory required.

Memory *location* refers to an address in the CPU's memory where a binary word can be stored. Each binary piece of data is a *bit* and eight bits make up one *byte*.

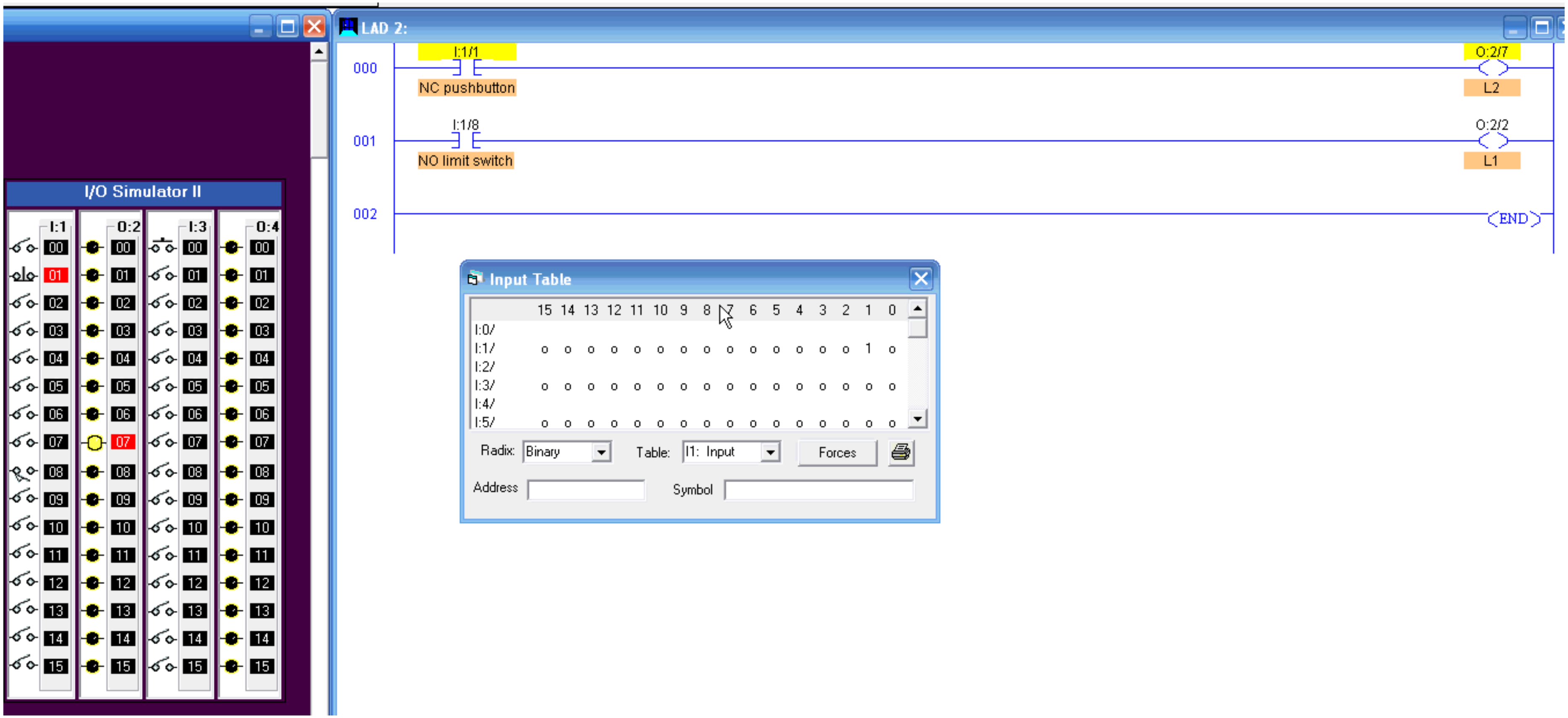


The program is stored in the memory as *1s* and *0s*, which are typically assembled in the form of *16-bit words.*

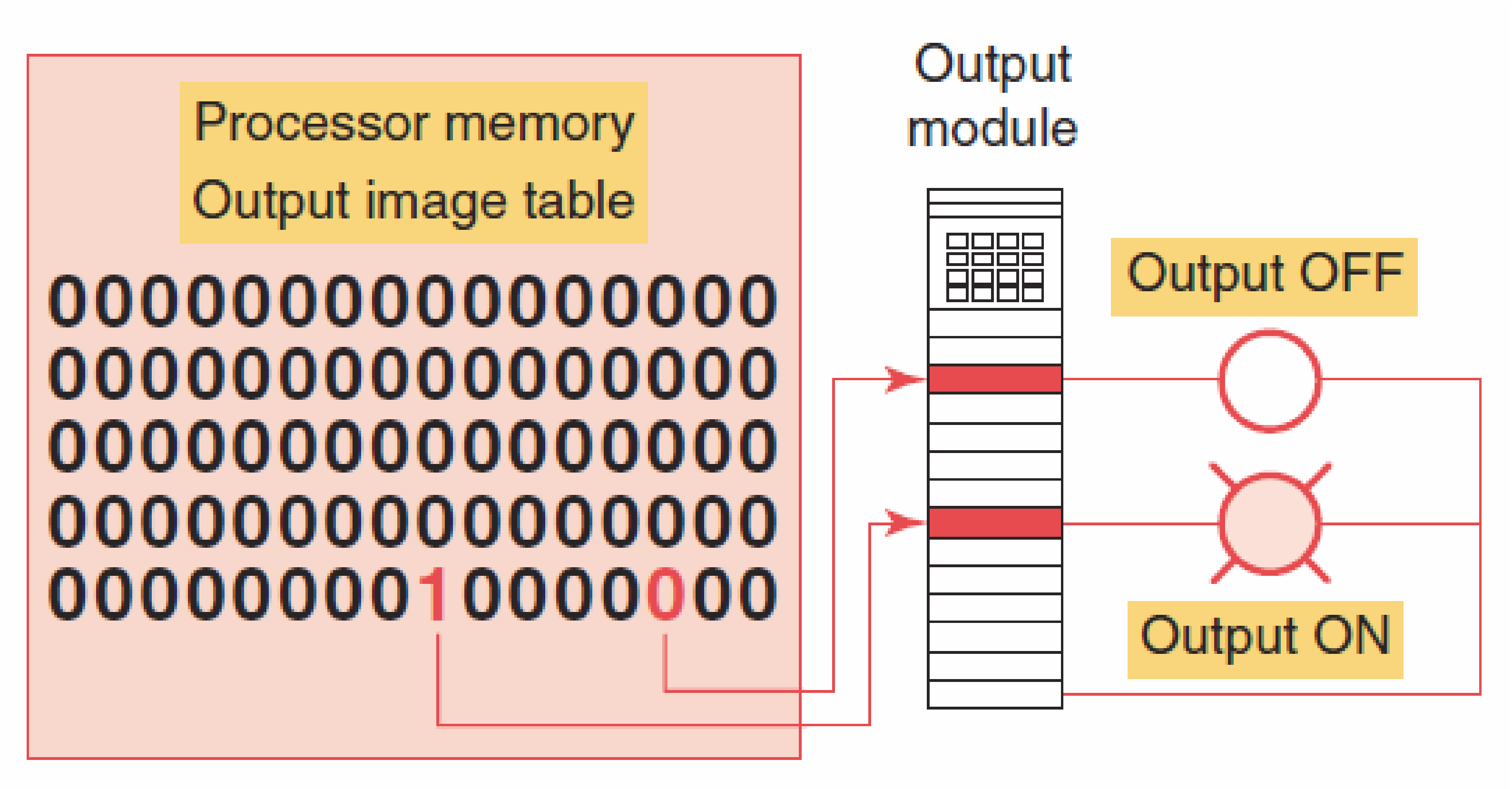
Sections of memory used to store the status of inputs are called *input status files or tables*.



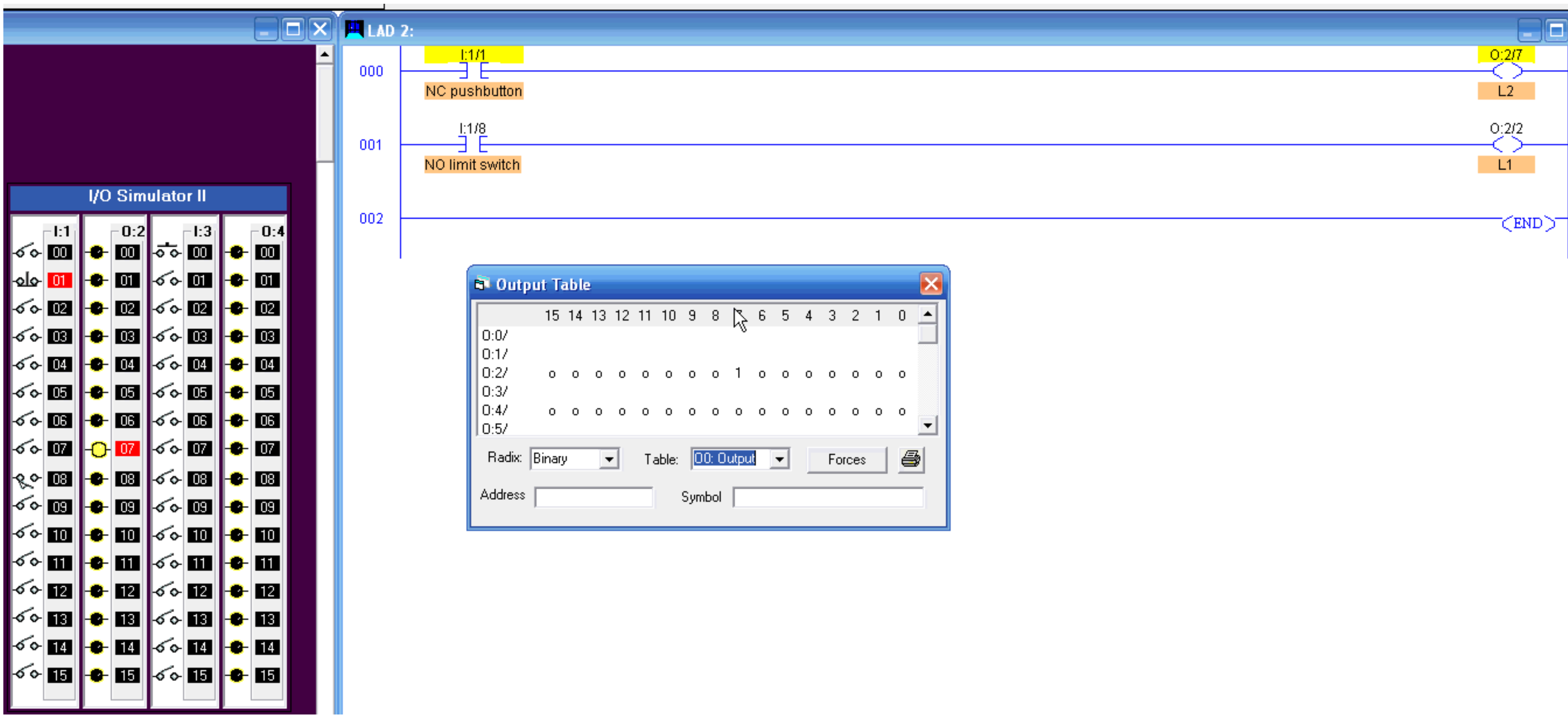
Input Table Simulation



Sections of memory used to store the status of outputs are called *output status files or tables*.



Output Table Simulation



2.8

Memory Types

Memory types can be placed into two general categories: *volatile* and *nonvolatile*.

Nonvolatile memory has the ability to retain stored information when power is removed accidentally or intentionally.

Volatile memory will lose its stored information if all operating power is lost or removed.

PLCs have programmable memory that allows users to develop and modify control programs. This memory is made nonvolatile so that if power is lost, the PLC holds its programming.

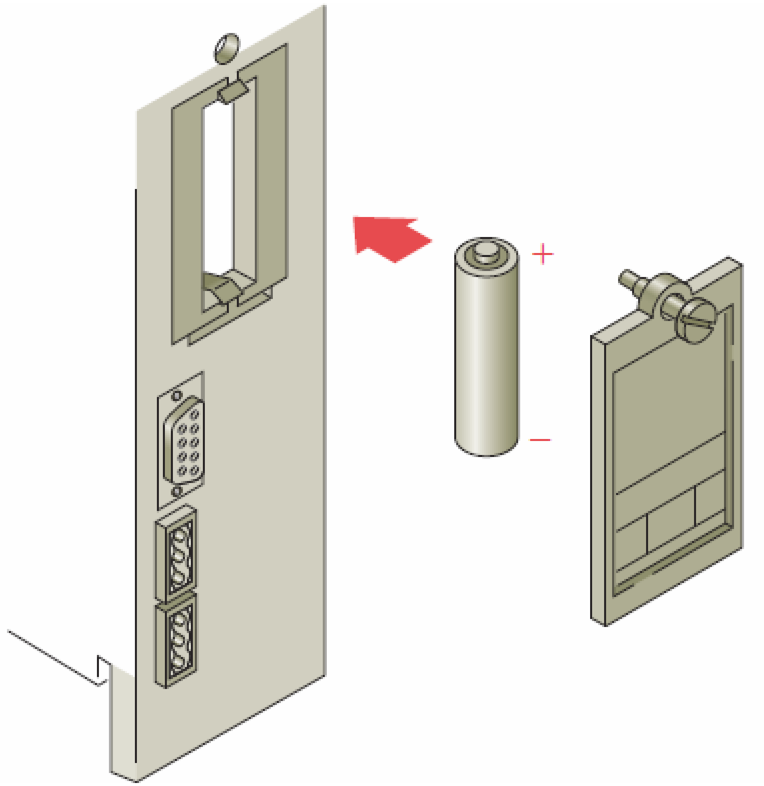
Nonvolatile *Read Only Memory (ROM)* stores programs, and data that cannot be changed after the memory chip has been manufactured.

ROM is used by the PLC for the operating system and controls the system software that the user uses to program the PLC.

Random Access Memory (RAM) **is designed so that information can be written into or read from the memory.**

PLCs use RAM as a temporary storage area of data that may need to be quickly changed.

RAM is volatile so battery backup is required for it to avoid losing data in the event of a power loss



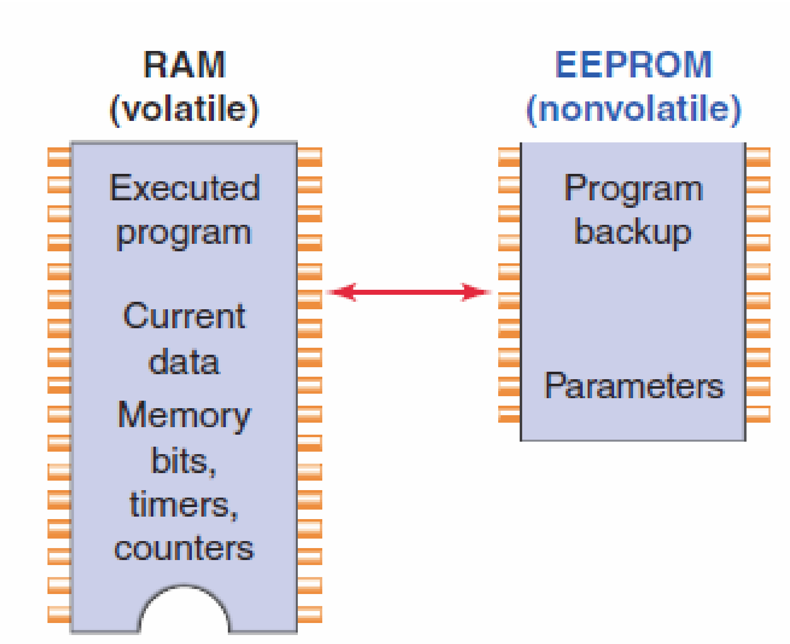
Erasable Programmable Read-Only Memory (EPROM) **provides some level of security against unauthorized or unwanted changes in a program.**

EPROMs are designed so that data stored in them can be read, but not easily altered without special equipment.

UV EPROM memory can only be erased with an ultraviolet light.

Electrically erasable programmable read-only memory (EEPROM) **is a nonvolatile memory that offers the same programming flexibility as does RAM.**

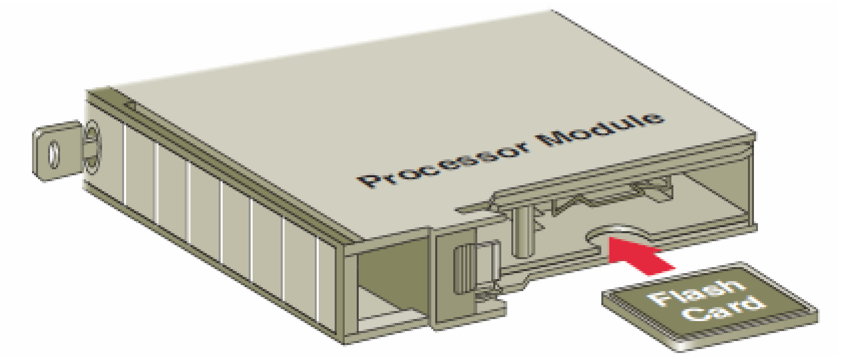
The EEPROM can be electrically overwritten with new data instead of being erased with ultraviolet light.



Because the EEPROM is nonvolatile memory, it does not require battery backup.

Flash EEPROMs **are similar to EEPROMs in that they can only be used for backup storage.**

Flash memory is extremely fast at saving and retrieving files.

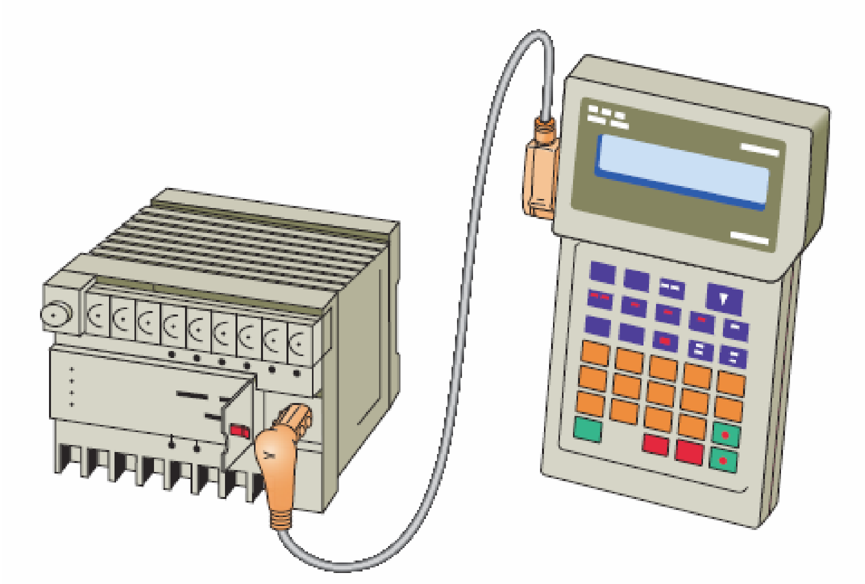


Flash memory is also sometimes built into the processor module, where it automatically backs up parts of RAM.

2.9

Programming Terminal Devices

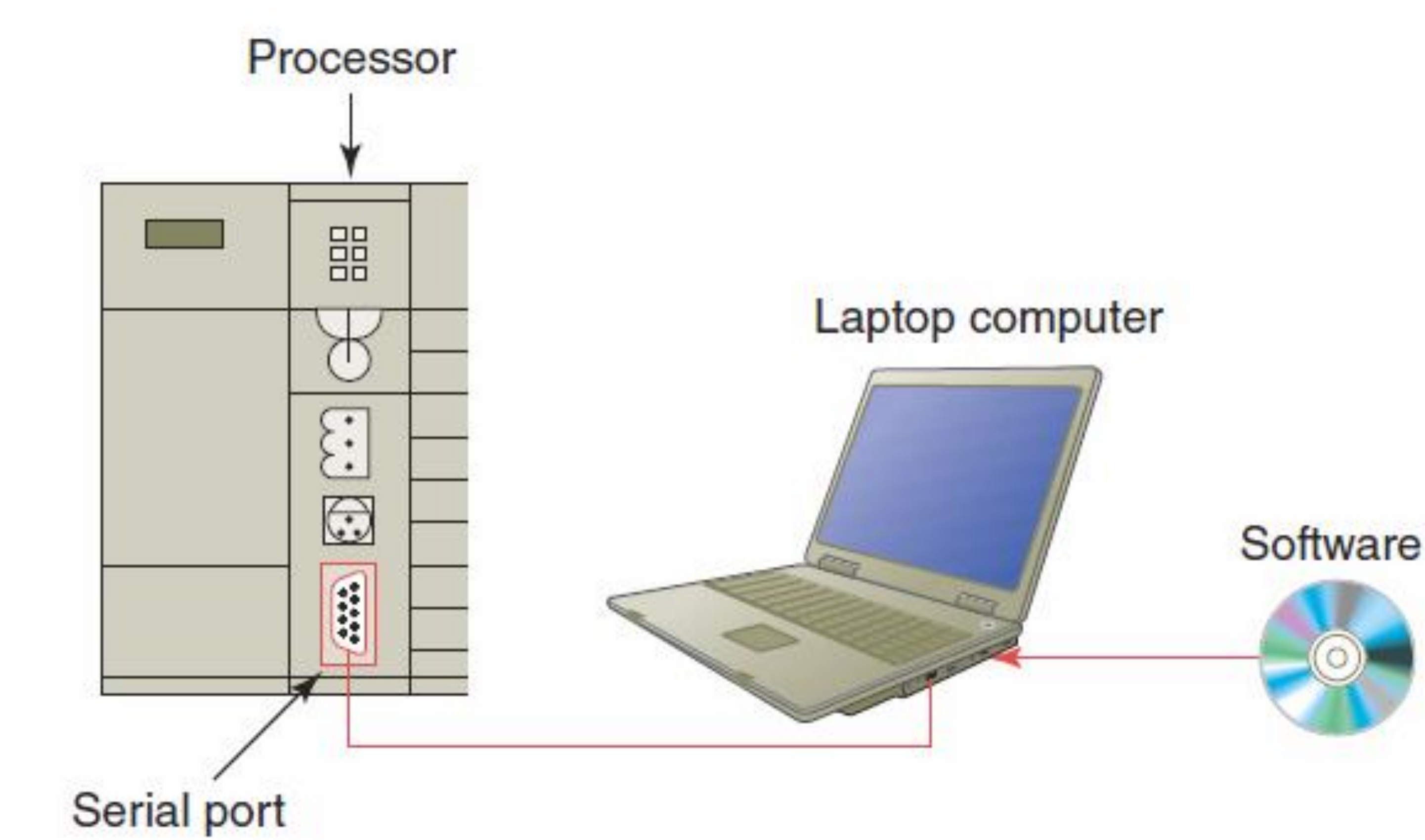
A *programming terminal* device is needed to enter, modify, and troubleshoot the PLC program.



The hand-held proprietary programming terminal has a connecting cable so that it can be plugged into a PLC's programming port.

Hand-held programmers are compact and inexpensive but have limited display capabilities.

The most popular method of PLC programming is to use a *personal computer* in conjunction with the manufacturer's programming software.

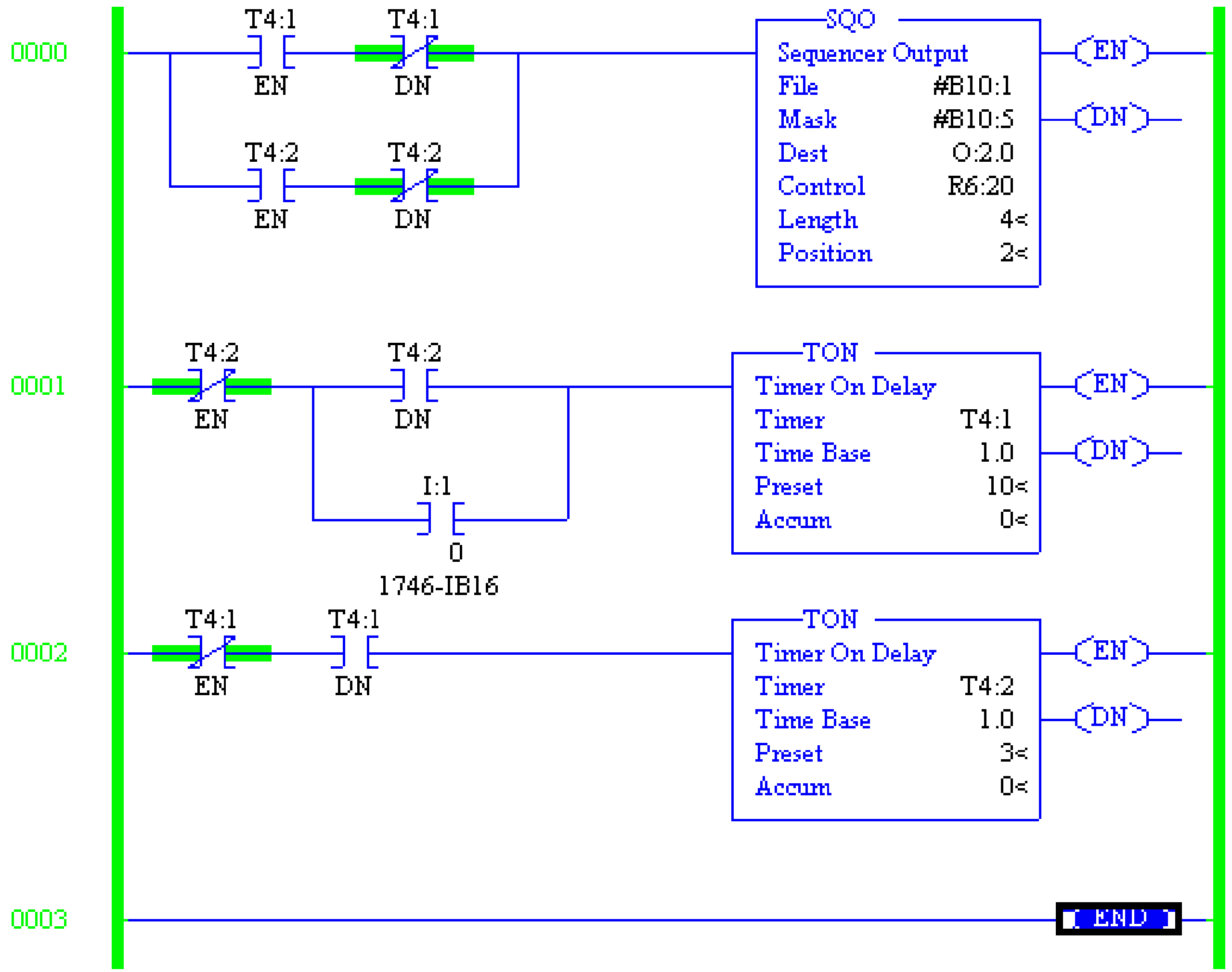


2.10

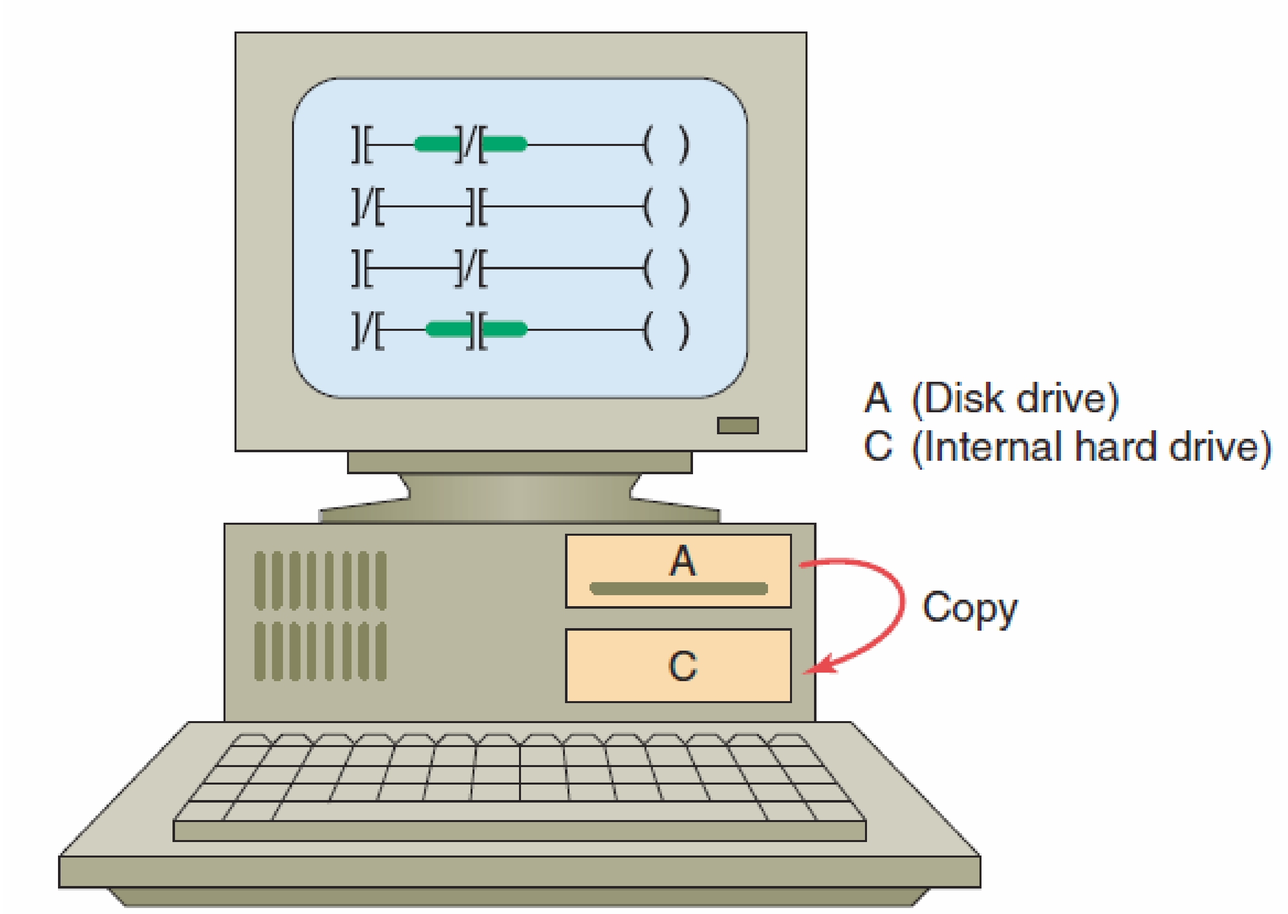
Recording and Retrieving Data

Printers are used to provide hard-copy printouts of the processor's memory in ladder program format.

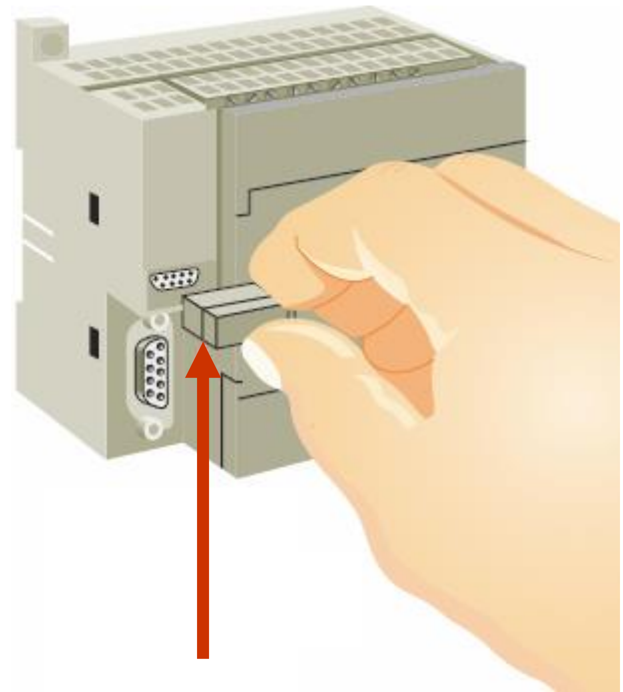
A **printout** can show programs of any length and analyze the complete program.



The program in the PLC is entered directly from the *keyboard* or *downloaded* from the computer hard drive or thumb drive.



Some CPUs support the use of a *memory cartridge* that provides portable EEPROM storage for the user program.



The cartridge can be used to copy a program from one PLC to another similar type PLC.

2.11

Human Machine Interfaces (HMIs)

A *human machine interface (HMI)* can be connected to communicate with a PLC and to replace pushbuttons, selector switches, pilot lights, thumbwheels, and other operator control panel devices

Human machine interfaces allow you to view the operation in real time.

You can configure display screens to:

- **Replace hardwired pushbuttons and pilot lights with realistic-looking icons.**
- **Allow the operator to change timer and counter presets.**
- **Show alarms, complete with time of occurrence.**